

COMPETITIVE MOTOR INSURANCE PRICING WITH SYNTHETIC QUOTE MARKETS: CONVERSION MODELLING AND PRICE OPTIMISATION

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Abstract

For an insurer, pricing the risk correctly is only half the problem. Competitive pricing requires insurers to balance expected claims costs with customer demand. However, the competing premiums and unsuccessful quotation outcomes needed to model customer behaviour are often unavailable in conventional insurance portfolio data. This dissertation develops a synthetic Australian motor insurance framework to investigate whether competitive market-position information can improve conversion modelling and pricing decisions. A portfolio of 50,000 synthetic policies is generated across five annual cohorts, combining technical pricing, a simulated market with six competing insurers, customer-choice modelling and price optimisation. Competitive information substantially improved conversion modelling. On the independent test sample, the selected Market-position GAM achieved a Spearman correlation of 0.817 with underlying conversion probabilities, compared with 0.623 for the Own-price GLM. Targeted repricing subsequently increased expected conversion from 18.28% to 21.91%, representing a relative improvement of 19.88%. This was achieved while keeping total quoted premiums close to the technical-pricing benchmark and increasing expected contribution after claims by 2.85%. Robustness analysis demonstrated that the predictive value of competitive information was consistent across the examined scenarios, although the financial benefits of repricing depended on customer price sensitivity. As customer behaviour and competitor pricing were simulated, these results should not be interpreted as estimates of outcomes achievable in the Australian motor insurance market. The principal contribution is a reproducible framework for developing and evaluating demand-aware insurance pricing methods when competitive quotation and conversion data are unavailable.

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1 Introduction

1.1 Motivation

In personal motor insurance, setting a premium that adequately reflects expected claims costs does not necessarily ensure that a customer will purchase the policy. Technical pricing estimates the cost of insuring a particular risk and provides the basis for calculating the premium offered to the customer. Commercial pricing must also consider the likelihood of purchase. Higher premiums may increase the contribution earned per policy but discourage customers from purchasing, while lower premiums may attract additional business at the expense of underwriting adequacy. Actuarial pricing therefore requires consideration of both expected claims costs and customer demand (Spedicato et al., 2018; Taylor, 1986).

This relationship is particularly important when customers compare quotations from multiple insurers. Price-comparison websites allow customers to assess competing offers within a single search, making an insurer's competitive position potentially important to the purchasing decision. A premium may be attractive when competing insurers charge more but comparatively expensive when several cheaper alternatives are available. Measures such as price rank and relative price provide information about this competitive position that cannot be determined from the insurer's premium alone. Regulatory investigations in Australia and the United Kingdom have examined the influence of comparison websites on insurance distribution and competition (Australian Competition and Consumer Commission, 2014; Financial Conduct Authority, 2014). Developments in automated purchasing may further increase the importance of comparing competing offers, although their implications for insurance customer behaviour remain uncertain (Deloitte, 2025).

A principal challenge is that the information required to model competitive customer demand may not be available alongside conventional insurance portfolio data. Insurers generally maintain records of the policies they underwrite, including customer characteristics, premiums, exposure and claims experience. These records support technical pricing but may not contain competitors' quotations or information about customers who requested a quote without purchasing. Consequently, an insurer may be able to estimate expected claims costs without having the information needed to understand how its competitive position influences conversion.

Existing actuarial research recognises the importance of customer price responsiveness and competition in determining commercially effective premiums (Guven and McPhail, 2013; Guelman and Guillén, 2014; Spedicato et al., 2018). However, examining these relationships is difficult when competing quotations and unsuccessful conversion outcomes are unavailable. The research problem addressed by this dissertation is therefore how to construct and evaluate a competitive pricing framework when the information required to estimate customer demand cannot be directly observed.

To investigate this problem, the study develops a synthetic Australian personal motor insurance environment informed by publicly available data. A synthetic portfolio and claims experience are generated to support technical pricing, followed by competing insurer quotations and customer purchasing decisions. The insurer whose pricing decisions are evaluated is referred to as the *reference insurer*, while six simulated competitors provide alternative quotations. The reference insurer's competitive position is measured using its premium relative to competing offers and its ranking among those offers. Conversion refers to a customer purchasing insurance from the reference insurer, with conversion probability representing the likelihood of that outcome.

The simulation specifies the underlying relationships governing claims and customer choice, collectively referred to as the *data-generating process (DGP)*. These relationships are known to the simulation but withheld from the models used to estimate claims costs, predict conversion and select pricing adjustments. Once the pricing strategy has been selected, its performance is evaluated using the underlying simulated customer-choice probabilities. This allows the study to examine whether improvements predicted by the fitted models also occur under the relationships used to generate customer behaviour.

The dissertation proceeds as follows. Section 2 reviews the literature on technical pricing, competition, customer demand and synthetic data. Section 3 describes the simulation, modelling and optimisation methodology. Section 4 presents the results, including predictive comparisons, pricing outcomes and robustness analyses. Section 5 discusses their implications and limitations, while Section 6 concludes the study and identifies opportunities for further research.

1.2 Research Questions

This dissertation is guided by two research questions.

- Can a synthetic motor-insurance framework informed by Australian public data generate a coherent quotation environment with the information required to approximate a comparison market?
- To what extent can competitive market-position information improve conversion modelling, and can the resulting demand model improve pricing decisions relative to a technical pricing benchmark?

The first question examines whether a synthetic dataset can establish consistent relationships between technical premiums, competing quotations, market position and customer conversion. The objective is to develop a controlled environment for investigating competitive pricing rather than to reproduce the actual pricing behaviour of Australian insurers and their customers.

The second question considers both predictive performance and pricing effectiveness. It examines whether competitive information improves conversion predictions beyond customer characteristics and the reference insurer's own premium, and whether the resulting models can support premium adjustments that increase expected conversion while maintaining specified financial requirements. This distinction is important because improved predictive accuracy does not necessarily lead to better pricing decisions.

1.3 Significance

The significance of this research lies in addressing the information gap between technical risk assessment and competitive pricing decisions. Although expected claims costs provide the actuarial basis for determining premiums, commercial outcomes also depend on whether customers accept the quotations offered. Where competitor prices and unsuccessful quotation outcomes are unavailable, the relationship between underwriting adequacy, competitive position and customer demand cannot readily be examined using conventional portfolio data.

The dissertation makes three methodological contributions. First, it develops a reproducible synthetic framework in which customer and vehicle characteristics, technical premiums, competing quotations and conversion outcomes are represented together. Competitive variables such as ranking and relative price are calculated directly from the generated quotations, ensuring that they reflect the available alternatives.

Second, the study distinguishes the value of additional competitive information from greater model complexity. Conversion models are compared using progressively richer information sets, followed by a comparison between a generalised linear model (GLM) and a generalised additive model (GAM) incorporating the same competitive information. This allows the study to investigate whether predictive improvements arise primarily from additional market information or from more flexible modelling methods.

Third, the framework evaluates conversion models through the pricing decisions they support. Premium adjustments are selected using estimated customer demand, subject to constraints designed to limit changes in the total premiums quoted across customers and preserve expected contribution after claims, defined as expected premium revenue less expected claims costs. The selected strategy is then evaluated using the underlying simulated customer-choice mechanism, providing an assessment beyond conventional predictive accuracy.

The practical contribution is a transparent and reproducible framework for investigating demand-aware insurance pricing when complete competitive quotation data are unavailable. Structural sensitivity, Monte Carlo regeneration and bootstrap analysis are used to examine the stability of the findings. The framework therefore provides a basis for developing and testing competitive-pricing methods, while recognising that application to observed insurance markets would require empirical validation of customer behaviour and competitor pricing.

2 Literature Review

2.1 Technical Pricing, Demand and Competition

2.1.1 Technical Adequacy and Commercial Pricing

Technical pricing forms the actuarial foundation of general insurance pricing. Its primary purpose is to estimate the expected cost of insuring a particular risk using policyholder and risk characteristics. This estimate provides a starting point for determining the premium offered to a customer, typically with additional allowances required for expenses, capital and target returns. However, a premium that adequately covers expected costs does not necessarily represent the most commercially effective price, as the insurer must also consider the likelihood that the customer will purchase the policy.

The distinction between technical and commercial pricing has long been recognised in actuarial literature. Taylor (1986) examines underwriting strategy in a competitive insurance environment, demonstrating the importance of considering market conditions alongside the insurer's own risk and financial objectives. More recent research by Spedicato et al. (2018) incorporates customer-demand modelling into pricing optimisation, recognising that commercial outcomes depend on both the premium charged and the probability that a policy is purchased.

This creates a trade-off between premiums and customer conversion. Increasing premiums may improve the contribution earned on each policy sold but discourage purchases, while lower premiums may attract additional customers at the expense of contribution per policy. Guven and McPhail (2013) examine price elasticity in insurance pricing, which describes how customer demand responds proportionately to changes in premiums. Understanding this relationship is important because changes in price may affect both the volume of business written and the financial contribution generated.

Guelman and Guillén (2014) further demonstrate the importance of distinguishing customer price responsiveness from other factors influencing purchasing behaviour. Their research highlights that customers receiving different premiums may also differ in risk characteristics and purchasing preferences, making the effect of price differences difficult to isolate from observed insurance data.

For the present dissertation, technical pricing establishes the risk-based benchmark against which alternative premiums are evaluated. Conversion modelling then provides the additional information needed to examine how pricing adjustments may affect customer demand.

2.1.2 Competitive Choice Sets and Quote Conversion

Customer conversion refers to the acceptance of an insurance quotation, usually represented as a binary outcome indicating whether the customer purchases the offered policy. In a competitive insurance market, customer conversion may depend on both the quoted premium and the alternative offers available. The collection of offers a customer can choose between is referred to as the *choice set*. An insurer's attractiveness within this set may depend on its premium relative to competing quotations, rather than its price in isolation.

Price-comparison websites provide a relevant example of this environment. They allow customers to compare premiums from multiple insurers within a single search, increasing price transparency and making differences between available offers easier to identify. The Australian Competition and Consumer Commission (2014) examines the influence of comparator websites on competition and consumer outcomes, while the Financial Conduct Authority (2014) investigates their role in the United Kingdom's general insurance market. These investigations establish the relevance of comparison platforms to insurance distribution, although they do not directly quantify the effect of individual competitive measures on conversion.

The importance of competitive position can be illustrated by considering an insurer offering a premium of $1,000. This may be the cheapest available quotation for one customer but substantially more expensive than the alternatives available to another. Although the insurer's premium is unchanged, its competitive position differs between the two customers. Measures such as price rank, which indicates the position of an offer when quotations are ordered from cheapest to most expensive, and relative price, which compares the offered premium with a competing price, can capture this difference.

These considerations motivate the inclusion of competitive market-position variables in conversion models. By generating alternative insurer quotations, the present study can examine whether information about the reference insurer's

position within each customer's choice set improves conversion prediction beyond customer characteristics and the reference insurer's own premium.

2.1.3 Strategic Pricing and Incomplete Market Observability

Actuarial research has examined how insurer pricing decisions interact with customer demand and competitive conditions. Taylor (1986) establishes an early framework for underwriting strategy under competition, while Emms et al. (2007) investigate optimal premiums when business volume responds to an insurer's price relative to the market. Pantelous and Passalidou (2013) similarly consider premium optimisation within a competitive insurance environment.

Further research incorporates strategic interactions between insurers. Game-theoretic models consider how one insurer's pricing decisions may influence, and be influenced by, the actions of competitors. Boonen et al. (2018) investigate these interactions in general insurance markets, while Mourdoukoutas et al. (2021) examine pricing in a competitive stochastic insurance setting. These approaches recognise that competing insurers may respond to changes in market conditions rather than maintaining fixed prices.

However, modelling strategic competition requires information or assumptions about competing insurers' objectives, pricing methods and likely responses. Such information may be difficult to obtain, particularly when insurers cannot observe competitors' internal pricing models or complete quotation histories.

The present dissertation adopts a narrower approach. Competing premiums are generated using specified pricing assumptions and held fixed while the reference insurer evaluates alternative prices. This allows changes in the reference insurer's competitive position and expected conversion to be examined without modelling how competitors might respond. The approach therefore represents a controlled pricing experiment rather than a full model of strategic competition.

Incomplete market information creates a related practical difficulty. Conventional insurance portfolio data may contain sufficient information to estimate expected claims costs but lack competitor quotations and outcomes for customers who did not purchase. The information required for technical pricing may therefore be available even when the information needed to model competitive demand is not. Constructing a synthetic quote market allows these components to be examined together under explicitly defined assumptions.

2.2 Conversion Modelling for Demand-Aware Pricing

2.2.1 Interpretable Benchmarks and Nonlinear Flexibility

Conversion modelling estimates the probability that a customer purchases an insurance policy using their characteristics, the quoted premium and other available information. Logistic regression provides a suitable starting point because conversion is a binary outcome and the model produces estimated purchase probabilities. As a generalised linear model (GLM), logistic regression also offers a relatively interpretable way of examining relationships between explanatory variables and customer purchasing behaviour.

However, customer responses to premiums and competitive position may be nonlinear. For example, moving from the second-cheapest quotation to the cheapest may have a different effect on conversion than moving between two relatively expensive quotations. A conventional logistic GLM assumes that its predictors have linear effects on the log-odds of conversion unless additional terms are introduced. Nonlinear relationships can be represented using transformations or splines, while generalised additive models (GAMs) allow selected predictors to have flexible, smooth effects estimated from the data.

Existing insurance research demonstrates the potential value of flexible modelling methods. Spedicato et al. (2018) compare conventional GLMs with machine-learning approaches in insurance price optimisation, illustrating how modelling choices can influence the resulting pricing decisions. Henckaerts et al. (2021) examine flexible methods for capturing complex relationships in insurance tariff modelling, while Yeo et al. (2001) investigate nonlinear customer responses to premium changes in motor insurance retention. Although retention differs from new-business conversion, these studies support examining whether flexible model structures can represent customer responses more effectively.

Greater model complexity does not necessarily produce better predictions. A flexible model may capture relationships specific to its training data without improving performance on new observations. Additional complexity may also make results more difficult to interpret and justify.

An important distinction for the present study is therefore whether conversion predictions improve because models incorporate more relevant competitive information or because they use greater statistical flexibility. To investigate this, logistic GLMs are compared using progressively richer information sets, beginning with technical-risk characteristics, followed by the reference insurer's own premium and then competitive market-position variables. The full Market-position GLM is subsequently compared with a GAM using the same competitive information. This separates the contribution of additional market information from the incremental benefit of nonlinear modelling.

2.2.2 Price Endogeneity

Price endogeneity arises when observed premiums are associated with other factors that also influence customer purchasing behaviour. In insurance, premiums are generally determined using policyholder and risk characteristics rather than assigned randomly. Consequently, an observed relationship between premiums and conversion does not necessarily identify how customers would respond to a change in price.

Consider two hypothetical customers. A 22-year-old driving a sports car receives a premium of \$1,800 and has a conversion probability of 15%, while a 45-year-old driving a family sedan receives a premium of \$900 and has a conversion probability of 30%. Although the customer offered the higher premium is less likely to purchase, the difference cannot be attributed to price alone. Age, vehicle type, income, purchasing preferences and other characteristics may also influence conversion.

The relevant question for pricing is how the same customer's probability of purchasing would change if their premium changed while other conditions remained fixed. For example, reducing the first customer's premium from \$1,800 to \$1,600 might increase their conversion probability from 15% to 22%. These probabilities are illustrative, but the comparison demonstrates the customer response that a pricing model aims to estimate.

In observed insurance data, only the purchasing outcome associated with the premium actually offered is generally available. The same customer's decision at an alternative premium cannot be observed simultaneously. This makes it difficult to distinguish the effect of changing price from differences between customers.

Guelman and Guillén (2014) address this identification problem when estimating price elasticity in automobile insurance using causal-inference methods. Similar difficulties arise in wider demand research, where observed prices and purchasing decisions may both depend on underlying consumer preferences and market conditions (Berry et al., 1995).

The synthetic framework used in this dissertation provides a controlled means of investigating alternative premiums. Customer purchasing outcomes are generated using a specified behavioural model, while separate conversion models attempt to estimate purchasing probabilities from the resulting observations. The fitted models are used to select premium adjustments, which are subsequently evaluated against the underlying customer-choice mechanism. This permits direct comparison of simulated outcomes under alternative premiums, although it does not establish causal price responses in an observed insurance market.

2.2.3 Prediction, Optimisation and Governance

Accurate conversion predictions do not necessarily lead to effective pricing decisions. A model may distinguish customers who are more likely to purchase insurance without accurately estimating how their purchasing probabilities change when premiums are adjusted. Consequently, improvements in measures such as log loss, which evaluates the accuracy of predicted probabilities, do not automatically demonstrate that a model will support better pricing decisions.

Mayorga and Torres (2017) examine automobile insurance pricing as an optimisation problem in which estimated customer responses inform premium selection. Related research by Emms et al. (2007) and Pantelous and Passalidou (2013) considers how expected business volume and financial outcomes interact when determining premiums. These studies demonstrate the importance of evaluating conversion models through the decisions they support, rather than relying solely on predictive accuracy.

For the present study, pricing optimisation involves selecting premium adjustments that increase expected conversion while satisfying specified financial constraints. These include limiting changes in total quoted premiums, preserving expected contribution after claims and maintaining an acceptable expected claims ratio. Expected contribution after claims represents expected premium revenue less expected claims costs, allowing for the probability that each customer purchases the policy. These measures are defined formally in Section 3.6.

Pricing decisions must also consider governance and customer outcomes. Models that assign different premiums across customer segments may produce results that are difficult to explain or justify, particularly where differences reflect sensitive characteristics or customer behaviour. The Australian Human Rights Commission and Actuaries Institute (2022) highlight the importance of avoiding unjustified discrimination in insurance pricing and underwriting, while the Australian Securities and Investments Commission (2024) examines governance challenges associated with artificial intelligence.

These considerations support using interpretable modelling methods, transparent assumptions and explicit financial constraints. The present study evaluates whether competitive information can improve pricing decisions within the simulated environment, but does not establish whether the resulting premiums would be appropriate for implementation. This would require additional assessment of expenses, regulation, fairness and customer outcomes.

2.3 Synthetic Data as an Experimental Framework

2.3.1 Synthetic Insurance Data and Incomplete Observability

Synthetic data consist of artificially generated observations designed to reproduce specified characteristics or statistical relationships. They are increasingly used where access to detailed information is restricted by privacy, confidentiality or commercial sensitivity. El Emam et al. (2020) discuss the generation of synthetic data for analytical applications, while Nowok et al. (2016) demonstrate methods for creating datasets that preserve selected statistical properties of observed information.

In actuarial research, synthetic data can support the development and evaluation of modelling methods without requiring access to confidential policyholder or claims records. Avanzi et al. (2021), for example, developed SynthETIC, an individual-claims simulator that generates insurance claims experience under controlled assumptions. Such approaches allow researchers to examine model performance under known conditions and systematically modify the relationships governing simulated outcomes.

The motivation for using synthetic data in this dissertation extends beyond confidentiality. The principal challenge is that the information required for competitive motor insurance pricing may not be available together in an accessible dataset. Portfolio data can support the estimation of expected claims costs, but conversion modelling also requires competing quotations and information about customers who did not purchase. A synthetic framework allows these components to be constructed and analysed jointly.

The present study uses Australian public data to inform selected demographic, geographic and vehicle characteristics. Claims experience, competing premiums and purchasing decisions are subsequently generated using specified assumptions. This produces a synthetic quote market in which technical risk, competitive position and customer conversion can be examined together.

However, calibrating selected portfolio characteristics to public data does not establish that the resulting market accurately represents Australian motor insurance. In particular, competitor pricing and customer-choice relationships are assumed rather than estimated from observed quotation behaviour.

The main methodological advantage is that the underlying relationships used to generate the data are known. In observed insurance data, individual conversion probabilities and responses to alternative premiums are generally unobservable. In the synthetic environment, these probabilities can be calculated directly from the customer-choice mechanism and used to evaluate fitted models and pricing strategies independently of the information available during their development.

2.3.2 Robustness

Evaluating the usefulness of synthetic data depends on the purpose for which they are generated. Conventional approaches commonly examine whether synthetic data can preserve the statistical characteristics of an original dataset, including variable distributions, dependencies and predictive relationships. These measures are useful when the objective is to create a representative substitute for observed data. However, they may be insufficient when the objective is to evaluate a modelling method or pricing decision.

Jordon et al. (2024) emphasise the importance of assessing synthetic data in relation to their intended application, while Jones et al. (2025) discuss their usefulness in actuarial research. These perspectives suggest that synthetic data

should be evaluated not only by how closely they resemble observed data, but also by whether they support reliable analysis of the questions being investigated.

This distinction is particularly important for competitive insurance pricing. A synthetic portfolio may contain plausible premiums, claims and customer characteristics without necessarily providing reliable conclusions about conversion modelling or pricing optimisation. The stability of those conclusions under alternative assumptions must therefore be considered.

The present dissertation addresses this through three complementary procedures. **Structural sensitivity analysis** examines how results change when assumptions about competition, customer demand or pricing constraints are altered. **Monte Carlo regeneration** repeats the full synthetic experiment using different random seeds to assess variation across independently generated datasets. **Paired bootstrap analysis** repeatedly resamples the independent test sample while holding the selected pricing strategy fixed, assessing the sensitivity of the final pricing comparison to the customers included.

Together, these procedures examine uncertainty arising from modelling assumptions, simulated observations and test-sample composition. They provide evidence about the internal reliability of the synthetic framework, while recognising that its applicability to real insurance markets cannot be established without observed quotation and customer-conversion data.

3 Synthetic Market Construction and Pricing Methodology

This study develops a fully synthetic Australian motor insurance framework to examine whether competitive market-position information improves conversion modelling and pricing decisions. The analysis proceeds through five stages. First, an insurance portfolio is generated using Australian public data and specified simulation assumptions. Second, claims experience is simulated and used to estimate technical premiums. Third, competing insurer quotations and customer purchasing decisions are generated. Fourth, conversion models are fitted using different combinations of customer, pricing and competitive information. Finally, the fitted models are used to select premium adjustments, which are evaluated against the underlying simulated customer-choice mechanism. Figure 1 summarises the modelling stages and the associated data splits.

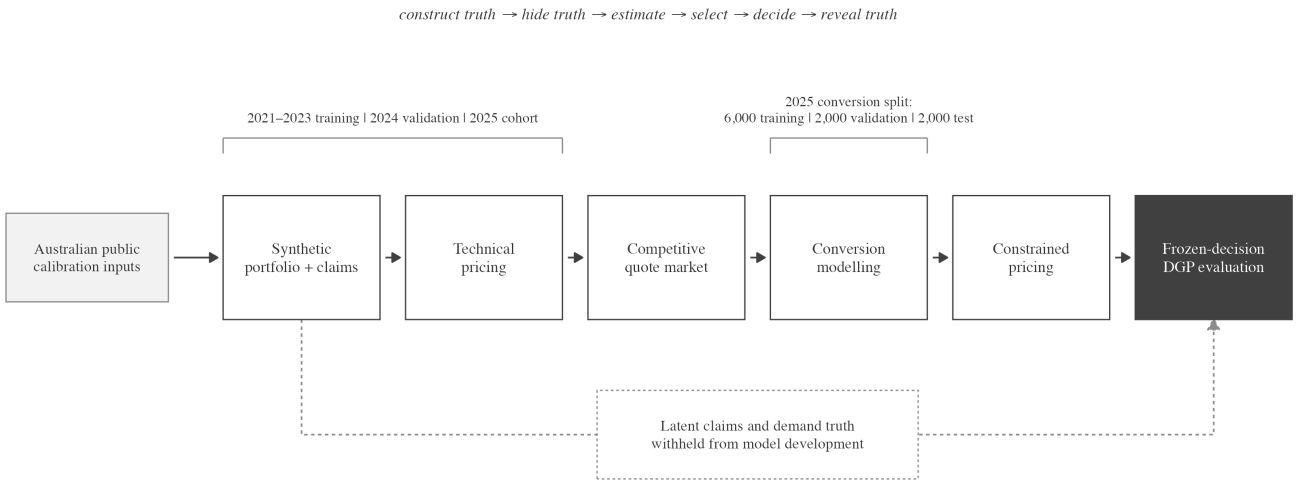


Figure 1: Overview of the synthetic motor insurance pricing framework, illustrating the sequence of modelling stages, the training and evaluation data splits, and the separation of underlying simulation outcomes from model development and final pricing evaluation.

3.1 Data and Empirical Calibration

The first stage constructs a synthetic portfolio of Australian comprehensive motor insurance policies. The portfolio uses publicly available data to inform the characteristics of individual policies, while variables for which suitable empirical information is unavailable are generated using explicit modelling assumptions.

This process, referred to as *empirical calibration*, involves converting published statistics and reference information into distributions and values from which synthetic observations can be generated. For example, vehicle registration data inform the composition of the simulated vehicle population, while published vehicle prices and depreciation estimates support the estimation of insured values.

The process is implemented in two modules. Module 00 prepares the calibration inputs, including vehicle registration distributions, geographic classifications and vehicle reference data. Module 01 uses these inputs to generate the individual policy records. Table 1 summarises the principal calibration sources. Further information on the corresponding assumptions and their implementation is provided in Appendix B.

Table 1: Principal empirical calibration sources and their role in the synthetic portfolio

Component	Principal source	Role in simulation
Vehicle registrations and fleet composition	ABS and BITRE	State, manufacture-year and make/model distributions
Driver population	BITRE licence-holder statistics	Driver-age distribution
Geography and remoteness	BITRE and ABS geographic and Census data	State and remoteness distributions
Vehicle segment and motive power	Green Vehicle Guide and FCAI	Vehicle classification and technology mix
Vehicle specifications	CarExpert historical records	Engine, power, torque, transmission and weight
Vehicle values	RACQ running-cost data	Reference prices and depreciation
Vehicle usage	ABS vehicle-use data	Motive-power usage relativities
Historical monetary values	ABS price indices	Adjustment of vehicle values and claims costs to a common price basis
Additional underwriting characteristics	Simulation assumptions	Excess, parking, financing and other policy attributes

3.1.1 Portfolio Structure

The portfolio consists of 50,000 synthetic comprehensive motor-insurance policies, equally distributed across five annual cohorts from 2021 to 2025. Each record represents an independent new-business opportunity with one year of coverage; cancellations and renewals are not modelled. The sample size provides sufficient observations for model estimation, validation and evaluation while keeping the simulation computationally manageable. Equal cohort sizes are a modelling choice and do not represent an observed insurer's annual business volumes.

The cohorts serve different purposes in technical-pricing development. Claims experience from 2021–2023 is used to fit candidate frequency and severity models. The 2024 cohort is used to compare their predictive performance and select the preferred specifications. Following selection, the models are subsequently refitted using the combined 2021–2024 experience and evaluated on the independent 2025 cohort.

The 2025 cohort also provides the 10,000 customer profiles used to construct the competitive quote market. These profiles are divided into 6,000 training observations, 2,000 validation observations and 2,000 test observations for conversion modelling. The training sample is used for initial model estimation, the validation sample supports model and pricing-strategy selection, and the test sample is reserved for final evaluation.

Six competing insurers are represented through simulated quotations. Their premiums are generated without constructing separate policyholder portfolios, claims histories or technical-pricing models for each competitor.

3.1.2 Generation of Policy Characteristics

The portfolio includes geographic, driver, vehicle and underwriting characteristics. These variables are generated in a defined sequence to preserve relevant dependencies. For example, licence experience depends on driver age, vehicle age depends on the manufacture year and policy year, and physical vehicle specifications are assigned using compatible reference records.

Geographic and driver characteristics: State is assigned using annual passenger-vehicle registration distributions published by ABS and BITRE. Remoteness is subsequently generated conditional on State using BITRE vehicle-location information and ABS geographic classifications. The five remoteness categories are Major Cities, Inner Regional, Outer Regional, Remote and Very Remote. Where postal areas overlap multiple remoteness categories, Census dwelling counts are used to allocate registrations between them (Bureau of Infrastructure and Transport Research Economics, 2025b; Australian Bureau of Statistics, 2021a, 2022).

Policies are then assigned to approximately 450 synthetic localities within the State and remoteness categories. Policies sharing a locality receive a common adjustment to expected claim frequency, representing geographic risk variation not explained by their other characteristics. These localities and their associated risk adjustments are simulation assumptions rather than observed postcode insurance experience.

Driver ages are generated using national licence-holder distributions published in the BITRE *Australian Infrastructure and Transport Statistics Yearbook 2025* (Bureau of Infrastructure and Transport Research Economics, 2025a). Ages are sampled within the corresponding age bands, and licence duration is derived from the simulated driver age and an assumed age at first licensing.

Vehicle characteristics:

Vehicle manufacture year is sampled from the relevant policy-year and State registration distribution. Vehicle age is then calculated as the difference between the policy year and manufacture year, with the simulation restricted to vehicles no more than 30 years old. Vehicle make, model, segment and motive power are assigned using the corresponding registration distributions and reference classifications.

Vehicle classifications are informed by the Australian Government's Green Vehicle Guide and FCAI segmentation criteria (Australian Government, 2026; Federal Chamber of Automotive Industries, nd). Additional vehicle specifications, including engine capacity, cylinder count, power, torque, transmission and unladen weight, are assigned using CarExpert vehicle records (CarExpert, 2026).

Vehicle values: Insured vehicle values are estimated using RACQ's 2025 vehicle-running-cost data, which provides a reference for new-vehicle prices and estimated values after five years (Royal Automobile Club of Queensland, 2025).

Each simulated vehicle is assigned a new-vehicle price and five-year depreciation ratio.

Let B_i denote the price assigned to a new-vehicle in 2025, r_i the proportion of that price remaining after five years and a_i the vehicle's age. The estimated vehicle value on the 2025 price basis is

$$V_i = \min\{B_i, \max(3000, B_i r_i^{a_i/5})\}$$

This extends the observed five-year depreciation ratio to other vehicle ages while imposing a minimum value of $3,000 and a maximum equal to the reference new-vehicle price. The extrapolated depreciation pattern and minimum value are modelling assumptions.

Historical monetary values are adjusted using ABS price indices. The motor-vehicle price index is used to express vehicle values and write-off amounts at their corresponding policy-year price levels, while the motor-vehicle maintenance and repair index is used for partial claims. Claims costs are subsequently converted to a common 2025 price basis for technical-pricing estimation. This prevents historical changes in vehicle and repair prices from being treated as differences in underlying claims risk (Australian Bureau of Statistics, 2026).

Annual usage and additional underwriting characteristics: Annual kilometres are generated from a lognormal distribution to allow for the positive skew typically associated with vehicle usage. The simulated values are adjusted for driver age, remoteness area and motive power, reflecting differences in expected driving patterns. Motive-power adjustments are informed by ABS vehicle-use data, while the age and remoteness adjustments are specified assumptions. Annual kilometres are scaled to approximately 11,100 km, based on the Australian Bureau of Statistics' average for passenger vehicles in the 12 months ended 30 June 2020 (Australian Bureau of Statistics, 2021b). Using this historical value as a common target for the 2021–2025 cohorts is a modelling assumption.

Additional characteristics include voluntary excess, overnight parking, vehicle financing, modifications, pre-existing damage and regular-driver status. Voluntary excesses are selected from $400, $600, $800, $1,000 and $1,500 using predefined probabilities. Their inclusion provides a more complete insurance record.

The completed portfolio provides the characteristics used to simulate claims and estimate technical premiums.

3.2 Synthetic Claims and Underlying Expected Claims Cost

Claims experience is generated using separate frequency and severity models. Frequency determines the number of claims reported during the policy year, while severity determines their costs and the payments made by the insurer. This approach is commonly used in general insurance because the factors influencing claim occurrence and claim size may differ (Frees, 2010; Ohlsson and Johansson, 2010).

The simulation specifies the underlying claims relationships before generating individual claim outcomes. Consequently, each policy has a known underlying expected annual claims cost. These expected costs are subsequently used to evaluate the fitted technical-pricing models.

3.2.1 Claim Frequency

Reported claim counts are generated using a Poisson distribution, with the expected number of claims determined by each policy's risk characteristics. For policy i, the reported claim count is generated as

$$N_i \mid \lambda_i \sim \text{Poisson}(\lambda_i),$$

where N_i is the number of reported claims and λ_i is the underlying expected annual claim frequency.

Expected frequency is calculated using multiplicative risk adjustments:

$$\lambda_i = c \times R_i^A \times R_i^L \times R_i^K \times G_{\ell(i)} \times U_i.$$

where c is a common frequency scaling factor, R_i^A, R_i^L and R_i^K represent the adjustments for driver age, licence experience and annual kilometres, respectively. The remaining terms introduce geographic and individual variation in claims risk.

The observable risk adjustments are specified as

$$\begin{aligned} R_i^A &= \min\left[2.20, \exp\{0.00055(A_i - 50)^2\}\right], \\ R_i^L &= \exp[-0.020 \min(L_i, 15)], \\ R_i^K &= \left(\frac{K_i}{11{,}100}\right)^{0.55}, \end{aligned}$$

where A_i is driver age, L_i is years licensed and K_i is annual kilometres.

These relationships allow claim frequency to vary with the characteristics of the insured driver. Expected frequency is lowest around age 50 and increases for younger and older drivers. The age adjustment is capped at 2.20 times its baseline value. Greater licence experience reduces expected frequency, with the adjustment capped after 15 years. Annual kilometres increase expected frequency at a diminishing rate, reflecting greater exposure to road use without assuming a proportional increase in claims.

Two additional components account for risk differences not captured by these variables. The geographic effect $G_\ell(i)$ is shared by policies within the same synthetic locality and follows a lognormal distribution with mean one and log-scale standard deviation 0.15. The policy-specific effect U_i follows a Gamma distribution with mean one and variance 0.20. These terms introduce additional variability among policies with otherwise similar observable characteristics (Dionne and Vanasse, 1989; Denuit et al., 2007).

The scaling factor c is calibrated so that the average probability of reporting at least one claim during 2025 is 14%, informed by the annual claimant-frequency reference reported by the Insurance Council of Australia (2025):

$$\frac{1}{n_{2025}} \sum_{i:\, \text{year}_i = 2025} [1 - \exp(-\lambda_i)] = 0.14.$$

Under the Poisson model, $1 - \exp(-\lambda_i)$ represents the probability of reporting at least one claim. The 14% target therefore concerns the proportion of policies expected to report a claim, rather than the average number of claims per policy.

The same underlying frequency is used to simulate current-year claims and five preceding years of claims history. These counts are generated independently conditional on the policy's expected frequency, allowing prior claim counts and consecutive claim-free years to reflect underlying risk differences while retaining random variation in reported experience.

3.2.2 Claim Severity

A separate claim amount is generated for each reported claim.

The simulation distinguishes between **write-off claims**, where the vehicle is written off, and **partial claims**, where damage does not result in a write-off.

Let T_{ij} indicate whether claim j for policy i is a write-off:

$$T_{ij} \sim \text{Bernoulli}(p_{WO}),$$

where p_{WO} is the probability that a reported claim results in a write-off. A common probability is applied across policies as a simplifying assumption.

For write-off claims, the gross claim amount is set equal to the simulated vehicle value. For partial claims, gross amounts are generated from a lognormal distribution:

$$X_{ij} \mid T_{ij} = 0 \sim \text{Lognormal}(\eta_i, \sigma_X^2),$$

where η_i and σ_X determine the location and variability of partial claim amounts on the logarithmic scale. The coefficient of variation is fixed at 1.00, allowing many relatively small claims alongside fewer large claims.

Expected partial claim severity also depends on vehicle value:

$$E(X_{ij} \mid T_{ij} = 0) \propto \left(\frac{V_i}{V_0}\right)^{0.35},$$

where V_i is the simulated vehicle value on the 2025 price basis and V_0 is the reference vehicle value. The exponent of 0.35 allows repair costs to increase with vehicle value without assuming a proportionate increase.

The insurer payment is determined after deducting the policyholder's voluntary excess:

$$Y_{ij} = \max(X_{ij} - e_i, 0),$$

where X_{ij} represents the gross claim amount and e_i is the excess. A reported claim may therefore result in no insurer payment if its gross amount does not exceed the excess.

The claims simulation is calibrated using two reference targets: an average insurer payment of approximately \$5,202 per reported claim and write-offs accounting for approximately 25% of total claims expenditure (Insurance Council of Australia, 2025). The write-off probability and the partial claim severity scale are calibrated jointly to achieve these targets after allowing for voluntary excesses.

The 25% assumption refers to the proportion of total claims expenditure attributable to write-offs, not the proportion of claims classified as write-offs. In the simulation, the \$5,202 target is calculated across all reported claims, including those resulting in no insurer payment.

Historical claim amounts are generated at the monetary values applicable to their policy year and then converted to the common 2025 price basis used for technical pricing.

The underlying expected annual claims cost for each policy is

$$\mu_i = \lambda_i s_i,$$

where λ_i is the expected annual frequency of reported claims and s_i is the expected insurer payment per reported claim, including the effect of voluntary excesses and claims resulting in zero payment.

Unlike realised claims experience, μ_i is calculated directly from the specified simulation relationships. It is retained for subsequent validation and is not used to fit the technical-pricing models.

3.3 Technical Pricing

Technical premiums are estimated using simulated claims experience and observable policy characteristics. The models follow the conventional frequency–severity approach, estimating annual claims costs from historical observations before applying a premium loading (Ohlsson and Johansson, 2010; Frees, 2010).

Frequency and severity modelling: Although the simulation generates all reported claims, the fitted technical-pricing models consider only claims resulting in an insurer payment. These are referred to as **paid claims**. Claims whose costs are fully covered by the policyholder's voluntary excess are excluded from claim frequency and severity, as they generate no insurer expenditure.

Let C_i denote the number of paid claims for policy i. Frequency is estimated using a quasi-Poisson GLM with a log link:

$$\log \mathbb{E}(C_i \mid X_i) = \beta_0 + X_i^\top \beta + \log E_i,$$

where X_i contains the observable risk characteristics, β represents the estimated regression coefficients and E_i is policy exposure. All policies have one year of exposure, so $E_i = 1$.

The quasi-Poisson model allows the variance of paid claim counts to differ from that assumed under a standard Poisson model. Its variance is specified as

$$\mathrm{Var}(C_i \mid \mathbf{X}_i) = \phi \mathbb{E}(C_i \mid \mathbf{X}_i),$$

where ϕ is the dispersion parameter. A value near one indicates variation close to that expected under a Poisson model, while a value above one indicates greater variation. This flexibility is relevant because the simulation includes geographic and individual risk differences not fully captured by the fitted predictors (Ver Hoef and Boveng, 2007).

Claim severity is estimated using a Gamma GLM with a log link:

$$\log E(S_i \mid X_i) = \gamma_0 + X_i^\top \gamma,$$

where S_i is the average insurer payment per paid claim and γ contains the estimated severity coefficients. The Gamma distribution accommodates positive, right-skewed payments, with variance increasing as expected severity increases.

All policies contribute to frequency estimation, including those with no paid claims. Severity estimation uses only policies with at least one paid claim. Each policy's average severity is weighted by its number of paid claims, so policies with multiple payments contribute proportionately to estimation (de Jong and Heller, 2008).

Model specification and selection: Four candidate specifications are considered separately for frequency and severity. These range from models using a smaller set of conventional risk characteristics to more flexible specifications incorporating nonlinear relationships and additional vehicle attributes. Natural cubic splines are used for selected continuous variables to capture relationships that may not be adequately represented by linear effects (Hastie et al., 2009).

The frequency candidates include driver age, licence experience, annual kilometres, claims history and geography, with additional specifications incorporating vehicle segment, motive power and physical vehicle characteristics. Severity candidates incorporate vehicle value, excess and vehicle segment, with more flexible models allowing nonlinear vehicle-value and vehicle-age effects and additional vehicle attributes. The complete candidate specifications are presented in Appendix C.

The candidate models are initially fitted using 2021–2023 experience and compared on the 2024 validation cohort. Frequency models are evaluated using mean Poisson deviance, while severity models are compared using claim-count-weighted mean Gamma deviance. Deviance measures predictive disagreement between the model and observed claims outcomes, with lower values indicating better fit on the validation data.

The preferred specification for each component is selected according to the lowest validation deviance. The selected models are then refitted using the combined 2021–2024 data. Multiplicative balancing adjustments align the predicted number of paid claims and their total payments with the observed development-period totals.

Calculation of the technical premium: The fitted expected annual claims cost for policy i is calculated as

$$\hat{\mu}_i = \hat{\lambda}_i \times \hat{s}_i,$$

where $\hat{\lambda}_i$ is the fitted annual frequency of paid claims and $\hat{s}_i$ is the fitted average insurer payment per paid claim.

A 30% loading is applied to estimated claims cost to determine the technical premium:

$$\pi_i = 1.30 \times \hat{\mu}_i.$$

This provides a simplified allowance for expenses, capital requirements and target returns. On the fitted claims-cost basis, the corresponding expected claims ratio is

$$\frac{1}{1.30} \times 100\% = 76.92\%$$

The loading is applied to estimated claims costs rather than to a premium already containing these additional allowances. The resulting technical premium provides the risk-based benchmark for the reference insurer and the common pricing reference used to generate competing quotations.

3.4 Competitive Quote Market and Conversion Generation

The 2025 cohort is used to construct a competitive quotation market. Each customer receives a premium from the reference insurer alongside quotations from competing insurers. These offers are used to calculate competitive-position measures, including relative price and price rank. Customer purchasing decisions are subsequently generated using a separate behavioural model that accounts for these competitive conditions.

3.4.1 Quote-Market Construction

Reference insurer premium: The reference insurer's initial quotation is generated by applying a commercial multiplier to its estimated technical premium:

$$P_i = \pi_i \omega_i, \qquad \omega_i \sim \text{Uniform}(0.95, 1.05)$$

where P_i is the reference insurer's initial quoted premium and ω_i is the commercial multiplier.

This allows initial quoted premiums to vary by up to 5% above or below technical pricing. The adjustment is generated independently of customer and vehicle characteristics, introducing price variation beyond that explained by estimated claims risk.

Competing insurer premiums: Six competing insurers are introduced, representing value-oriented, mainstream, premium and specialist pricing approaches. Their quotations are generated relative to the reference insurer's fitted technical premium, with adjustments for each competitor's pricing level and preference for particular customer groups.

For competitor k and customer i,

$$Q_{ik} = \pi_i \exp(\alpha_k + \delta_{ik} + \varepsilon_{ik}),$$

where Q_{ik} is the competing premium, α_k represents the competitor's general pricing level, δ_{ik} represents its pricing adjustment for the customer's segment and ε_{ik} introduces random quote variation.

These adjustments allow different insurers to offer different prices for the same customer. For example, the young-driver specialist applies more favourable adjustments to younger drivers, while the low-risk specialist is relatively more competitive for customers with favourable claims histories.

Random quote variation follows a normal distribution with mean zero and standard deviation 0.045 on the logarithmic price scale. Competing premium multipliers are restricted to between 0.75 and 1.35 times the technical premium to limit extreme quotations.

Each competitor also has a specified probability of returning a quotation, reflecting the assumption that not every insurer provides an offer for every customer. If fewer than two competitors initially return a quote, the two competitors with the smallest availability draws are retained. Each customer therefore receives between two and six competing quotations. The baseline competitor parameters and segment adjustments are documented in Appendix B.

Competitive market position: The reference insurer's competitive position is calculated using the quotations returned for each customer.

Let $\mathcal{K}_i$ denote the set of competitors providing a quotation to customer i. The cheapest competing premium is

$$Q_i^{\min} = \min_{k \in \mathcal{K}_i} Q_{ik}.$$

The reference insurer's relative price is then

$$RP_i = \frac{P_i}{Q_i^{\min}}.$$

A relative price of 1.10, for example, indicates that the reference insurer's premium is 10% higher than the cheapest competing quotation. A value below one indicates that the reference insurer is cheaper than every available competitor.

Its price rank is

$$R_i = 1 + \sum_{k \in \mathcal{K}_i} I(Q_{ik} < P_i),$$

where $I(Q_{ik} < P_i) = 1$ when competitor k is cheaper, and zero otherwise; hence the number of indicators equal to one will reveal how many competing insurers are priced below the reference insurer.

These measures are calculated directly from the generated quotations rather than assigned independently. As a result, they can be recalculated whenever the reference insurer's premium changes during pricing optimisation.

3.4.2 Two-Stage Conversion Data-Generating Process

Customer purchasing decisions are generated using a two-stage choice model. The first stage determines whether the customer purchases insurance through the simulated comparison market. The second determines which insurer is selected, conditional on a purchase occurring.

This distinction is important: **market participation** represents purchasing from any available insurer, whereas **reference insurer conversion** represents purchasing specifically from the reference insurer.

Stage 1 - Market participation: The probability that customer i purchases insurance from any insurer in the available choice set is modelled using a logistic function:

$$q_i = \text{logit}^{-1}\left(\theta_0 + \mathbf{Z}_i^\top \theta\right),$$

where q_i is the probability of purchasing through the market, Z_i contains the characteristics influencing the purchasing decision and θ represents their effects.

Market participation depends on the median premium within the customer's available quotations relative to the overall market median. Higher relative premiums reduce the probability of purchase, while driver age and previous claims introduce additional variation.

The intercept is calibrated to produce an average market participation probability of 78% across the 2025 cohort. This is an assumed simulation target.

Stage 2 - Insurer choice: Conditional on purchasing insurance, customers are assigned probabilities of selecting each available insurer through a random-utility model. In this framework, utility represents the relative attractiveness of an insurer's offer to an individual customer (Train, 2009).

For insurer h in customer i's choice set, utility is specified as

$$U_{ih} = B_{ih} + \tau_{ih} - \beta_i \log(r_{ih}) + H(R_{ih}) - Q(r_{ih}) - K(R_{ih}) + \kappa I(h = \text{ref}).$$

The terms represent different influences on customer choice:

- B_{ih} captures the customer's non-price preference for an insurer.
- τ_ih introduces additional variation in preferences between customers and insurers.
- β_i measures customer-specific price sensitivity.
- r_{ih} compares an insurer's premium with the cheapest offer across the complete available choice set.
- $H(R_{ih})$ provides additional preference for highly ranked quotations.
- $Q(r_{ih})$ penalises quotations that exceed a specified relative-price threshold.
- $K(R_{ih})$ penalises quotations falling below a specified ranking threshold.
- κ adjusts the reference insurer's utility to achieve the intended average conversion rate.

The price-sensitivity and preference terms allow customers with similar observable characteristics to respond differently to the same quotations. The ranking and threshold components also permit nonlinear responses when an insurer moves between competitive positions. Their baseline parameter values are provided in Appendix B.

Utilities are converted to insurer-choice probabilities using a multinomial logit function:

$$w_{ih} = \frac{\exp U_{ih}}{\sum_{\ell \in \mathcal{J}_i} \exp U_{i\ell}},$$

where $\mathcal{J}_i$ is the complete set of available insurers, including the reference insurer. The resulting conditional probabilities sum to one across the choice set.

The reference insurer's underlying conversion probability is

$$p_i = q_i \times w_i,$$

where w_i is the conditional probability of selecting the reference insurer after deciding to purchase through the market.

The reference insurer's utility adjustment is calibrated to achieve an average initial expected conversion rate of 18%. This is distinct from the 78% market participation target because it concerns purchases from the reference insurer alone.

Generation of conversion outcomes: Customer purchase outcomes are simulated from the two-stage model. Each customer first receives a purchase decision based on their market participation probability. Customers who purchase are then assigned an insurer according to their conditional choice probabilities.

The reference insurer's conversion outcome equals one if the customer purchases its policy and zero otherwise. A zero includes both customers who select a competing insurer and those who do not purchase through the market.

This produces 10,000 binary conversion observations for the subsequent modelling stage. The underlying conversion probabilities remain separately available for final evaluation but are not supplied as outcomes to the fitted conversion models.

3.5 Conversion Modelling

Conversion models are fitted to estimate the probability that a customer purchases insurance from the reference insurer using observable characteristics, quoted premiums and competitive market information.

Six candidate models are compared to determine whether incorporating information about competing quotations improves predictions beyond customer characteristics and the reference insurer's own premium.

3.5.1 Model Specification

Five logistic GLMs and one GAM are considered. The GLMs examine the value of different information sets, while the GAM tests whether additional nonlinear flexibility improves predictions when the full competitive information set is available.

Table 2: Candidate conversion models and their explanatory information

Model	Information included
Technical-only GLM	Customer and vehicle characteristics, claims history and estimated technical premium
Own-price GLM	Technical-only variables plus the reference insurer's quoted premium relative to technical premium
Rank GLM	Own-price variables plus the reference insurer's rank percentile
Relative-price GLM	Own-price variables plus its premium relative to the cheapest competitor
Market-position GLM	Combines rank and relative price with additional price-gap, competitor-price variation and quote-availability measures
Market-position GAM	Uses the same broad competitive information as the Market-position GLM, with additional nonlinear flexibility

The Technical-only GLM provides the initial benchmark using customer characteristics and estimated claims risk.

The Own-price GLM introduces the reference insurer's commercial pricing adjustment, measured by

$$OP_i = \log\left(\frac{P_i}{\pi_i}\right),$$

where P_i is the initial quotation and π_i is the fitted technical premium. This measures the quoted premium relative to the risk-based benchmark, rather than treating the quoted premium as independent of expected claims costs.

The Rank and Relative-price GLMs provide separate extensions of the Own-price model. The former introduces competitive rank, adjusted for the number of available quotations, while the latter introduces the reference insurer's price relative to the cheapest competitor.

The full Market-position GLM combines these measures with additional information, including the price gap above a market threshold, the reference premium relative to the market median, competitor-price dispersion and the number of competing quotations. These variables describe different aspects of the customer's available offers. The detailed information sets and specifications are provided in Appendix C.

Natural cubic splines are included for selected continuous variables to accommodate nonlinear relationships. The Market-position GAM introduces further flexibility through penalised cubic regression smooths, with smoothing parameters estimated using restricted maximum likelihood.

Comparing the Market-position GLM and GAM therefore examines the incremental benefit of greater model flexibility while retaining the same broad competitive information.

3.5.2 Model Validation and Selection

The 10,000 quote profiles are divided into 6,000 training observations, 2,000 validation observations and 2,000 test observations. The candidate models are initially fitted using the training sample and compared using the validation sample.

The primary model-selection measure is binary **log loss**, which evaluates the accuracy of predicted conversion probabilities:

$$LL = -\frac{1}{n}\sum_{i=1}^{n}\left[D_i \log(\hat{p}_i) + (1 - D_i)\log(1 - \hat{p}_i)\right],$$

where D_i is the observed conversion outcome and $\widehat{p_i}$ is its fitted conversion probability.

Lower log loss indicates better probability predictions. This is particularly relevant because pricing optimisation uses predicted purchase probabilities rather than classifications of whether individual customers will purchase.

The Market-position GLM is the default specification for the full competitive information set. The GAM is selected only if it reduces validation log loss by at least 0.5% relative to the GLM and achieves a calibration slope between 0.70 and 1.30.

The **calibration slope** assesses whether the spread of predicted probabilities is appropriate. A value near one indicates that predicted probabilities vary between customers at approximately the appropriate scale, while values substantially below one generally indicate predictions that are too extreme. The selection rule therefore requires the GAM to demonstrate additional predictive value without producing unacceptable calibration.

Following model selection, the specifications are refitted using the combined training and validation observations and assessed on the independent test sample. The underlying simulated conversion probabilities are then used to evaluate prediction accuracy through mean absolute error, root mean squared error and Spearman rank correlation. Spearman correlation measures how well predicted probabilities preserve the ordering of customers by their underlying conversion probabilities.

The pricing optimisation uses a related but distinct development process. Conversion models initially fitted on the training sample are recalibrated using validation outcomes, and premium adjustments are selected using the validation sample. Following selection, the strategy is evaluated on the independent test sample. Test observations are not used to determine the model specification, probability recalibration or pricing decisions.

This separation provides an independent evaluation of the selected strategy, while recognising that validation data contribute to both recalibration and pricing selection.

3.6 Pricing Optimisation

The conversion models are used to identify premium adjustments that improve expected customer conversion while satisfying specified financial constraints. This reflects the trade-off between attracting customers through more competitive prices and preserving sufficient premiums to cover expected claims costs (Spedicato et al., 2018; Mayorga and Torres, 2017).

Four strategies are compared: the technical-pricing benchmark, uniform repricing using the Own-price GLM, uniform repricing using the selected Market-position model, and targeted repricing using the selected Market-position model.

The uniform strategies apply one multiplier to every customer's technical premium. The targeted strategy allows different multipliers across customer segments and forms the principal pricing experiment.

3.6.1 Candidate Prices and Probability Calibration

Alternative premiums are generated by applying candidate multipliers to the fitted technical premium:

$$P_i(m_i) = m_i \pi_i, \qquad m_i \in \{0.95, 0.96, \ldots, 1.05\}.$$

where m_i is the proposed premium multiplier and π_i is the fitted technical premium. A multiplier of 1.00 represents the technical-pricing benchmark, while 0.95 and 1.05 represent reductions and increases of 5%, respectively.

These bounds limit the size of premium adjustments and reduce the extent of extrapolation beyond the commercial pricing variation introduced during quote-market generation. Unlike the initial reference insurer quotation in Section 3.4, candidate prices are defined directly relative to technical premium to provide a consistent benchmark across strategies.

For every candidate price, competitive-position variables are recalculated using the proposed reference insurer premium and unchanged competing quotations. This includes its rank, price relative to competing offers and additional market-position measures. Predicted conversion probabilities are then obtained using the updated variables.

Before premium adjustments are selected, the conversion models fitted on the training sample undergo probability recalibration using validation outcomes:

$$\operatorname{logit}(p_i^{\text{cal}}) = a + b \operatorname{logit}(\widehat{p}_i),$$

where $\widehat{p}_i$ is the original predicted conversion probability, a adjusts the overall probability level and b adjusts the spread of predictions.

Predicted conversion probabilities are recalibrated using observed validation outcomes before being used for pricing optimisation. This adjusts systematic differences between predicted and observed conversion rates, improving the reliability of the probabilities used to estimate customer demand and expected contribution after claims. Recalibration is performed separately for the Own-price GLM and the selected Market-position model, as their prediction errors may differ. The resulting adjustments are then applied consistently across candidate premiums, without using test-sample outcomes.

3.6.2 Targeted Segmentation

The targeted strategy assigns a common premium multiplier to customers within each selected segment. Candidate segmentation variables include driver characteristics, claims history, geography, vehicle characteristics, annual kilometres and voluntary excess.

Continuous characteristics are generally divided into quartiles using cut-offs determined from the validation sample. Categorical characteristics retain their relevant groups. A minimum of 50 validation observations is required per segment to avoid selecting premiums using very small customer groups.

Each candidate segmentation variable is considered separately rather than combining multiple characteristics into highly detailed pricing cells. For example, vehicle-value segmentation allows the optimiser to select a different premium multiplier for each vehicle-value quartile.

The multipliers are selected using predicted conversion and financial outcomes. No predetermined rule requires a particular group to receive a discount or premium increase.

Each candidate segmentation structure is assessed on the validation sample. The preferred structure is selected according to its model-expected conversion, provided that it satisfies the pricing constraints described below.

Where candidate structures achieve expected conversion rates within 0.05 percentage points of the highest result, the structure with fewer segments is preferred. This avoids introducing additional segmentation complexity for negligible predicted improvements.

3.6.3 Optimisation

For each candidate segmentation, premium multipliers are selected using binary linear programming, implemented through the R package `lpSolve`.

Binary linear programming selects from a set of predefined alternatives while enforcing constraints. In this application, the optimiser chooses one premium multiplier for each customer segment.

Let x_{gm} equal one when multiplier m is selected for segment g, and zero otherwise. Exactly one multiplier is selected for each customer group:

$$\sum_m x_{gm} = 1 \qquad \text{for every segment } g.$$

The objective is to maximise the total predicted number of customer conversions:

$$\max_x \sum_g \sum_m x_{gm} \widehat{N}_{gm},$$

where $\widehat{N}_{gm}$ is the predicted number of conversions within segment g under multiplier m.

The optimisation is subject to constraints intended to preserve the overall quoted premium level and specified measures of claims adequacy and financial contribution.

Table 3: Pricing constraints applied to the targeted repricing

Constraint	Requirement	Purpose
Premium adjustment	0.95–1.05	Limit the size of individual premium changes
Aggregate quoted rate level	99.5%–100.5%	Maintain approximate rate neutrality
Contribution retention (all customers)	At least 100%	Preserve expected contribution after claims
Contribution retention (each customer group)	At least 90%	Limit underwriting deterioration within individual segments
Expected claims ratio	At most 80%	Maintain claims adequacy
Expected conversion	At least 100% of benchmark	Prevent reductions in expected conversion

Aggregate quoted rate level: This measures the total premiums that would be quoted under a strategy relative to the total premiums quoted at technical rates:

$$RL = \frac{\sum_i P_i(m_i)}{\sum_i \pi_i}, \qquad 0.995 \le RL \le 1.005.$$

A value of 100% indicates that the total quoted premium remains unchanged relative to the technical-pricing benchmark. The permitted range limits the aggregate change to 0.5% in either direction.

The calculation includes all quotations, irrespective of whether a customer purchases. It therefore measures the quoted premium level, not premium revenue expected from accepted policies.

Expected contribution after claims: This measures the expected premium revenue from customers who purchase insurance, less their fitted expected claims costs. For each customer, the difference between the proposed premium and expected claims cost is weighted by the predicted probability of purchase:

$$\widehat{M} = \sum_i \hat{p}_i(P_i(m_i))[P_i(m_i) - \hat{\mu}_i].$$

Here, $\hat{p}_i(P_i(m_i))$ is the calibrated conversion probability at the proposed premium, and $\hat{\mu}_i$ is the fitted expected annual claims cost.

This measure accounts for both the contribution per accepted policy and the number of policies expected to be purchased. It excludes separately modelled operating expenses, acquisition costs and capital charges, and therefore does not represent net insurer profit.

The strategy must preserve at least the benchmark expected contribution after claims across all customers. These requirements limit financial deterioration in individual groups while allowing the strategy to improve conversion overall.

Contribution retention compares expected contribution after claims under the proposed strategy with the technical-pricing benchmark:

$$\text{Contribution retention} = \widehat{M}_{\text{strategy}}/\widehat{M}_{\text{benchmark}}.$$

Note: Contribution retention is reported as a percentage, so a value of 100% indicates no change from the benchmark.

The targeted strategy must retain at least 100% of benchmark contribution across the portfolio and at least 90% within each selected segment. These requirements limit the financial deterioration permitted in exchange for greater conversion.

Expected claims ratio: This measures expected claims expenditure as a proportion of expected written premium. Both amounts account for the probability that each quoted policy is purchased:

$$CR = \frac{\sum_i \hat{p}_i\big(P_i(m_i)\big)\widehat{\mu}_i}{\sum_i \hat{p}_i\big(P_i(m_i)\big)P_i(m_i)} \leq 0.80.$$

The maximum claims ratio of 80% is a specified modelling constraint rather than a regulatory requirement. The technical-pricing benchmark has a fitted-basis claims ratio of approximately 76.92%, so the ceiling permits some deterioration in claims adequacy while restricting excessive premium reductions.

The targeted strategy must also achieve at least the technical-pricing benchmark's model-expected conversion. Together, these constraints allow premiums to be redistributed between customer groups while limiting aggregate premium changes and preserving the specified financial requirements.

Uniform-pricing comparisons: Two additional strategies are evaluated using the same multiplier to every customer's technical premium: one based on the Own-price GLM and the other on the selected Market-position model.

Unlike targeted repricing, these strategies maximise model-expected contribution after claims, subject to retaining at least 95% of benchmark conversion and satisfying portfolio contribution and claims-ratio constraints. They are not subject to the narrow aggregate quoted rate-level restriction.

The uniform strategies therefore serve as supplementary comparisons rather than directly equivalent rate-neutral alternatives to targeted repricing.

3.6.4 Evaluation

Pricing decisions are selected using the validation sample before the chosen strategy is applied to the independent test sample. The selected segmentation, premium multipliers and fitted probability-recalibration adjustments remain fixed.

Competing quotations are also held unchanged when evaluating alternative reference insurer prices for a given customer.

The final evaluation uses the underlying customer-choice mechanism introduced in Section 3.4. Rather than relying solely on the conversion probabilities predicted by the fitted models, the simulation recalculates the probability of purchasing from the reference insurer under each selected premium.

This provides an independent assessment of the strategy against the specified customer-choice relationships.

For strategy s, mean expected conversion is

$$\overline{p}_s = \frac{1}{n}\sum_{i=1}^{n} p_i(s),$$

where $p_i(s)$ is the underlying simulated probability of purchasing from the reference insurer under strategy s.

Absolute and relative improvements are calculated as

$$\Delta p_s = \overline{p}_s - \overline{p}_0$$

$$\Delta p_s^{rel} = \frac{\overline{p}_s - \overline{p}_0}{\overline{p}_0},$$

where $\overline{p}_0$ is expected conversion under the technical-pricing benchmark.

Absolute conversion uplift is expressed in percentage points, while **relative conversion uplift** measures the proportional increase from the benchmark. For example, an increase from 18% to 21% represents an absolute improvement of three percentage points and a relative improvement of approximately 16.7%.

Expected contribution after claims is also recalculated using the underlying customer-choice probabilities:

$$M_s = \sum_i p_i(s) \left[P_i(s) - \hat{\mu}_i \right].$$

This calculation uses the underlying simulated purchase probabilities but retains the fitted estimates of expected claims costs. The resulting financial measure therefore evaluates the selected demand strategy using estimated actuarial costs rather than providing a complete assessment of insurer profitability.

Final performance is assessed using expected conversion, absolute and relative conversion uplift, aggregate quoted rate level, contribution retention and expected claims ratio.

These measures establish whether the pricing adjustments selected using fitted customer-demand models remain effective when evaluated against the underlying simulated customer-choice mechanism.

3.7 Validation, Sensitivity and Uncertainty

Additional validation and robustness procedures assess whether the principal findings remain consistent under alternative assumptions, independently generated synthetic datasets and changes in the composition of the final test sample.

Validation and information controls: The modelling framework maintains a separation between model development and final evaluation.

The 2025 cohort is excluded from technical-pricing model estimation. For conversion modelling and pricing optimisation, the independent test sample is withheld from model specification, probability recalibration and strategy selection.

The underlying expected claims costs and customer-choice probabilities are also stored separately from the observations used to fit models and select premiums. They are used for subsequent evaluation, rather than being supplied directly to the fitted models.

Validation checks are performed throughout the computational framework to assess the consistency of generated records, model predictions and pricing calculations. In particular, competitive-position variables are recalculated for each candidate premium to ensure that predicted conversion reflects the revised quotation environment.

Robustness analysis: Three complementary procedures are used to examine different sources of uncertainty, as summarised in Table 4.

Table 4: Robustness procedures and sources of uncertainty

Procedure	What changes?	Purpose
Structural sensitivity analysis	Market, demand and pricing assumptions	Assess dependence on the specified modelling assumptions
Monte Carlo regeneration	Randomly generated portfolios, claims and conversion outcomes	Assess stability across independent synthetic datasets
Paired test-sample bootstrap	Composition of the held-out test sample	Quantify sampling uncertainty in the final pricing comparison

Structural sensitivity analysis: Ten cases are evaluated, comprising the baseline and nine alternatives covering competitive conditions, customer demand and pricing constraints.

Market scenarios examine stronger or weaker competition and greater variation in competing premiums. Demand scenarios vary customer price sensitivity, ranking preferences and nonlinear price thresholds. Pricing scenarios alter the permitted range of premium adjustments or impose a stricter claims-ratio ceiling.

Each scenario changes a related group of assumptions while retaining the remaining baseline structure. The detailed parameter changes are documented in Appendix B.

Only the affected stages of the simulation are rerun. Market scenarios regenerate competing quotations and subsequent modelling stages. Demand scenarios retain the original quotations but regenerate customer-choice outcomes and subsequent models. Pricing scenarios retain the fitted market and demand structure while repeating optimisation under revised constraints.

The underlying portfolio and technical-pricing models remain unchanged across these scenarios. Where possible, common random-number seeds are retained to reduce variation unrelated to the assumptions being examined.

Each scenario is evaluated according to whether competitive market-position information improves conversion modelling beyond the Own-price GLM and whether targeted repricing increases expected conversion while preserving at least the benchmark expected contribution after claims.

Monte Carlo regeneration: This procedure examines whether the findings depend on the particular random observations generated in the baseline experiment.

The complete simulation and modelling process is repeated using ten independent sets of random-number seeds. Empirical calibration inputs and structural assumptions remain fixed, while new portfolios, claims experience, competing quotations and conversion outcomes are generated.

Technical-pricing and conversion models are refitted, and model and pricing-strategy selection are repeated using the same procedures as in the baseline analysis.

This allows predictive performance, selected model specifications, targeting structures and pricing outcomes to be compared across independently generated datasets. Given the limited number of replications, the analysis is interpreted as a stability assessment rather than a precise estimate of simulation uncertainty.

Paired bootstrap analysis: The final procedure examines uncertainty associated with the particular customers included in the 2,000-profile test sample.

A paired nonparametric bootstrap is conducted using 5,000 samples drawn with replacement (Efron and Tibshirani, 1993). For each resample, the technical-pricing benchmark and selected targeted strategy are evaluated using the same customer profiles.

Keeping the observations paired preserves the relationship between each customer's expected outcomes under the two strategies, allowing uncertainty in their difference to be assessed.

The fitted models and selected pricing decisions remain fixed throughout resampling. Each bootstrap sample is used to recalculate expected conversion uplift, contribution retention, expected claims ratio and aggregate quoted rate level.

Two-sided 95% bootstrap percentile intervals are calculated using the 2.5th and 97.5th percentiles of the resulting distributions.

Together, the three procedures examine sensitivity to modelling assumptions, variation across independently generated datasets and uncertainty associated with the composition of the final test sample. They assess the internal reliability of the framework, while its applicability to observed insurance markets remains conditional on the assumptions underlying the simulated data.

3.8 Computational Implementation

The analysis is implemented in R through a sequence of R Markdown modules. Module 00 prepares empirical calibration inputs, Modules 01–06 implement the core simulation, modelling and optimisation framework, Module 07 performs structural sensitivity analysis and Module 08 conducts Monte Carlo regeneration. A supplementary Module 06B implements the paired test-sample bootstrap, while Module 09 produces dissertation figures and tables.

Intermediate datasets, model metadata and validation checks are written explicitly between stages rather than retained in an undocumented computing environment. Fixed random-number seeds support reproduction of the baseline experiment, while replication-specific seeds are used for Monte Carlo analysis.

The accompanying computational materials and appendices document the modelling specifications, simulation assumptions, required calibration inputs and supplementary results. The main text therefore describes the methodology and its underlying assumptions without reproducing implementation code line by line.

4 Results

This section presents the results of the synthetic motor insurance framework. First, the technical-pricing models and competitive quote market are examined to assess whether the simulation produces a coherent environment for analysing customer conversion. The predictive performance of the conversion models is then compared to determine the value of incorporating competitive market-position information. Finally, pricing optimisation evaluates whether these improvements support more effective premium decisions, while structural sensitivity, Monte Carlo regeneration and bootstrap analysis assess the stability of the findings.

4.1 Technical Pricing and Synthetic Market Validation

4.1.1 Technical Pricing

The synthetic portfolio comprised 50,000 policies and generated 7,765 reported claims, of which 7,306 resulted in insurer payments. The remaining 459 claims were fully covered by policyholder excesses and therefore did not contribute to the claim frequency or severity used for technical pricing.

Validation on the 2024 cohort favoured a quasi-Poisson frequency GLM incorporating nonlinear effects of driver age, licence experience and annual kilometres, alongside claims history, geography, vehicle segment and motive power. For severity, a Gamma GLM incorporating nonlinear effects of vehicle value and age, together with voluntary excess and vehicle segment, achieved the strongest validation performance. Including additional physical vehicle specifications did not improve the selected models' predictive performance, and these variables were therefore excluded from the final specifications.

The selected models were subsequently refitted using combined 2021–2024 experience and evaluated on the independent 2025 cohort.

Model performance was assessed using several complementary measures. **Observed-to-expected (O/E) ratios** compare actual claims experience with model predictions. A ratio of one indicates agreement, while values below or above one indicate that the model overpredicted or underpredicted the observed outcome, respectively. Dispersion estimates describe the variability remaining in the fitted models, while Spearman correlation assesses how well fitted claims costs preserve the ranking of policies according to their underlying expected risk. Mean absolute error (MAE) measures the average absolute difference between fitted and underlying expected claims costs, whereas root mean squared error (RMSE) places greater weight on larger errors.

Table 5 summarises the model-selection results and validation diagnostics.

Table 5: Technical-pricing model selection and validation diagnostics

Diagnostic	Result
Selected frequency model	Quasi-Poisson GLM with nonlinear effects and motive power
Selected severity model	Gamma GLM with nonlinear effects of vehicle value and age
Frequency dispersion	1.023
Severity dispersion	1.231
2025 frequency O/E	0.974
2025 severity O/E	0.947
2025 claims-cost O/E	0.947
Underlying claims-cost Spearman correlation	0.802
Mean fitted-to-underlying claims-cost ratio	1.018
Claims-cost RMSE	$440.67
Claims-cost MAE	$282.46

The fitted frequency and severity models produced dispersion estimates of 1.023 and 1.231, respectively. The 2025 claims-cost O/E ratio was 0.947, indicating that realised claims costs were approximately 5.3% below model predictions. The models therefore provided reasonably close aggregate estimates, although they modestly overestimated the observed claims experience.

The synthetic framework also permits comparison with the underlying expected claims costs specified during data generation. As illustrated in Figure 2, fitted and underlying expected claims costs exhibited a Spearman correlation of 0.802. This indicates that the models recovered substantial differences in expected risk between policies, despite not having access to the underlying claims relationships during estimation.

Mean fitted claims cost was 1.8% above its underlying expectation. However, the RMSE of $440.67 and MAE of $282.46 indicate that estimation errors remained at the individual-policy level. The stronger aggregate agreement therefore does not imply that every policy's expected claims cost was estimated accurately.

Across all five annual cohorts, fitted expected claims cost averaged $820.64 per policy. Applying the 30% loading produced a mean technical premium of $1,066.83. By comparison, the 2025 cohort used for the competitive quote market had a mean technical premium of $1,070.43.

The fitted premiums therefore provided a risk-differentiated pricing benchmark while retaining the estimation uncertainty that would arise when technical premiums are developed from observed claims experience rather than known underlying claims costs.

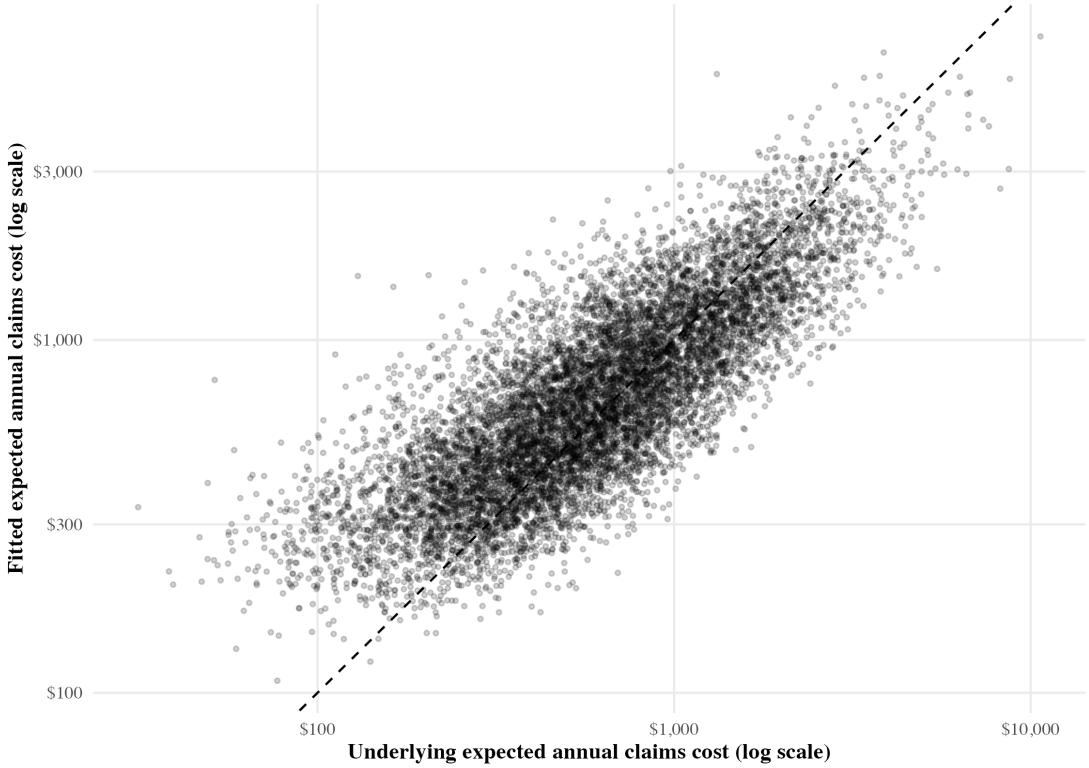


Figure 2: Recovery of underlying expected claims cost by the fitted technical-pricing model on the 2025 cohort.

4.1.2 Competitive Quote Market and Conversion Environment

The 2025 cohort was used to construct the competitive quote market. Each customer received between two and six competing quotations, with an average of 5.64 competitors returning an offer. Table 6 summarises the resulting market characteristics.

Table 6: Baseline characteristics of the synthetic 2025 quote market

Quote-market measure	Result
Customer profiles	10,000
Mean number of returned competitors	5.64
Mean technical premium	$1,070.43
Mean reference insurer quote	$1,070.47
Mean competing insurer quote	$1,075.87
Median reference insurer rank	4
Mean reference insurer rank	3.75
Reference insurer cheapest	4.57%
Reference insurer within 5% of cheapest	29.49%
Median relative price to cheapest	1.0766
Mean gap above cheapest	7.85%

Although the reference insurer's average premium was close to the average competing premium, this did not translate into a favourable position within individual customer comparisons.

The reference insurer's median price rank was fourth, and it offered the cheapest quotation in only 4.57% of cases. Its median relative price was 1.0766, meaning that its quotation was typically 7.66% above the cheapest available competing offer.

This distinction illustrates why average premiums across all customers provide an incomplete measure of competitiveness. An insurer may charge premiums close to the market average while being undercut by at least one competitor in most individual comparisons. The generated market therefore provides meaningful variation in competitive position despite the similarity in average quoted premiums.

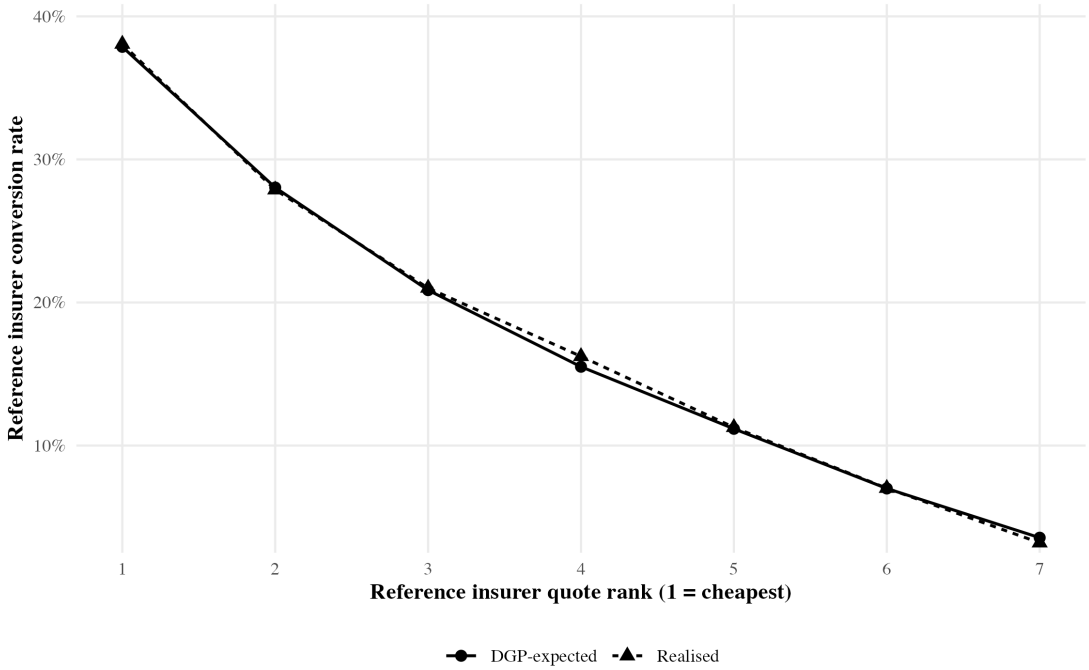
The simulated purchasing outcomes also closely matched their calibration targets. Across the 2025 cohort, the average underlying probability of purchasing insurance from any available insurer was 78.00%, compared with a realised participation rate of 78.04%. For the reference insurer specifically, underlying expected conversion was 18.00%, compared with realised conversion of 18.02%.

These are distinct measures: market participation includes purchases from any available insurer, whereas reference insurer conversion counts purchases from the reference insurer alone.

Customer conversion varied substantially with competitive position. As shown in Figure 3, the reference insurer's underlying expected conversion was highest when it offered the cheapest quotation, reaching 38.38%, and declined as its ranking deteriorated.

The Spearman correlations between underlying conversion probability and the reference insurer's price rank and relative price were -0.718 and -0.722, respectively. The negative relationships indicate that customers were less likely to select the reference insurer when it ranked less favourably or charged more relative to its competitors.

These findings confirm that the generated quotation environment contains the competitive-demand relationships specified by the simulation. They therefore establish a suitable setting for examining whether conversion models can recover those relationships from observable quotation data.



The relationship is generated by the synthetic DGP and is not an empirical estimate of Australian customer behaviour.

Figure 3: Underlying expected and realised reference insurer conversion by competitive quote rank.

4.2 Conversion Model Performance

4.2.1 Validation Fit and Model Selection

Six conversion models were compared to assess the predictive value of customer characteristics, the reference insurer's own price and competitive market-position information.

Performance was primarily evaluated using **log loss**, which measures the accuracy of predicted conversion probabilities, with lower values indicating better predictions. The area under the receiver operating characteristic curve (**AUC**) measures how well a model distinguishes customers who purchase from those who do not. An AUC of 0.5 indicates discrimination no better than chance, while an AUC of one indicates perfect discrimination.

The **calibration slope** assesses whether predicted probabilities vary between customers at an appropriate scale. A value near one is desirable, while a value substantially below one generally indicates that predictions are too extreme.

Table 7 presents the validation performance of each model. It also reports Spearman correlations with the underlying simulated conversion probabilities on the independent test sample. These correlations were calculated after model selection and were not used to select the preferred specification.

Table 7: Conversion-model validation performance and recovery of underlying conversion probabilities

Model	Validation log loss	Validation AUC	Validation calibration slope	Test underlying Spearman
Technical-only GLM	0.4645	0.5205	0.293	0.150
Own-price GLM	0.4462	0.6364	0.967	0.623
Rank GLM	0.4387	0.6641	0.971	0.708
Relative-price GLM	0.4378	0.6647	1.016	0.746
Market-position GLM	0.4307	0.6850	0.995	0.812
Market-position GAM	0.4286	0.6889	1.062	0.817

The Technical-only GLM performed poorly, with validation log loss of 0.4645 and AUC of 0.521. Its AUC was only slightly above 0.5, indicating limited ability to distinguish customers who purchased from those who did not. The calibration slope of 0.293 also indicated substantial differences between the spread of predicted probabilities and that supported by the observed outcomes.

Introducing the reference insurer's own quoted price improved validation log loss to 0.4462, a reduction of 3.94%, while AUC increased to 0.636. This demonstrates that own-price information contributed predictive value beyond customer characteristics and technical risk estimates.

Further improvements occurred when competitive information was included. The Rank and Relative-price GLMs reduced validation log loss to 0.4387 and 0.4378, respectively. The full Market-position GLM performed better still, achieving log loss of 0.4307, AUC of 0.685 and a calibration slope of 0.995.

These results indicate that the reference insurer's position relative to competing quotations contains information not captured by its own premium alone.

The Market-position GAM achieved the lowest validation log loss of 0.4286 and the highest AUC of 0.689. Its relative log-loss improvement over the Market-position GLM was 0.51%, narrowly exceeding the predefined 0.5% model-selection threshold. The GAM also satisfied the calibration-slope requirement, with a value of 1.062, and was therefore selected for the baseline analysis.

However, the additional benefit of the GAM was considerably smaller than the improvement achieved by incorporating competitive information. Relative to the Technical-only GLM, the Market-position GLM reduced validation log loss by approximately 7.28%, whereas the GAM provided a further reduction of only 0.51%.

The results therefore suggest that richer competitive information contributed more to predictive performance than the additional nonlinear flexibility introduced by the GAM.

4.2.2 Test Performance and Recovery of Underlying Conversion Probabilities

Following model selection, the Market-position GAM was refitted using the combined training and validation observations and evaluated on the independent test sample.

The model achieved test log loss of 0.4433 and AUC of 0.6679, compared with validation values of 0.4286 and 0.6889. Although performance declined moderately on the independent sample, the model continued to distinguish customers with different conversion propensities.

The highest predicted probability decile recorded a realised conversion rate of 33.0%, compared with 18.05% across the test sample. Customers ranked in this decile were therefore approximately 1.83 times as likely to convert as the average customer in the test sample.

The model's mean predicted conversion was 18.51%, compared with realised conversion of 18.05%. Its calibration slope was 0.926, while expected calibration error was 0.0176.

Expected calibration error summarises the average difference between predicted probabilities and realised conversion rates across groups of customers with similar predictions. The results indicate reasonably close agreement between predicted and observed conversion, although some calibration error remained.

The synthetic framework also allows predictions to be compared directly with the underlying conversion probabilities specified by the customer-choice mechanism. The GAM's mean predicted conversion of 18.51% was close to the underlying expected rate of 18.65%.

However, agreement across the full test sample does not necessarily imply accurate predictions for individual customers. The probability MAE was 0.0498 and RMSE was 0.0651, equivalent to average errors of approximately 4.98 percentage points and an RMSE of 6.51 percentage points, respectively.

The selected GAM achieved a Spearman correlation of 0.817 with the underlying conversion probabilities, compared with 0.812 for the Market-position GLM and 0.623 for the Own-price GLM. This indicates that both full market-position models recovered substantially more of the variation in underlying conversion probabilities than the model using own-price information without competitive position.

The small difference between the Market-position GLM and GAM further supports the finding that competitive information provided a larger improvement than additional nonlinear model flexibility.

These results also illustrate the distinction between predicting individual purchasing outcomes and estimating their underlying probabilities. Individual purchases contain random variation, so even a well-specified probability model cannot identify every customer who will purchase. Strong agreement with the underlying conversion probabilities provides additional evidence that the selected model recovered important features of the simulated customer-choice process.

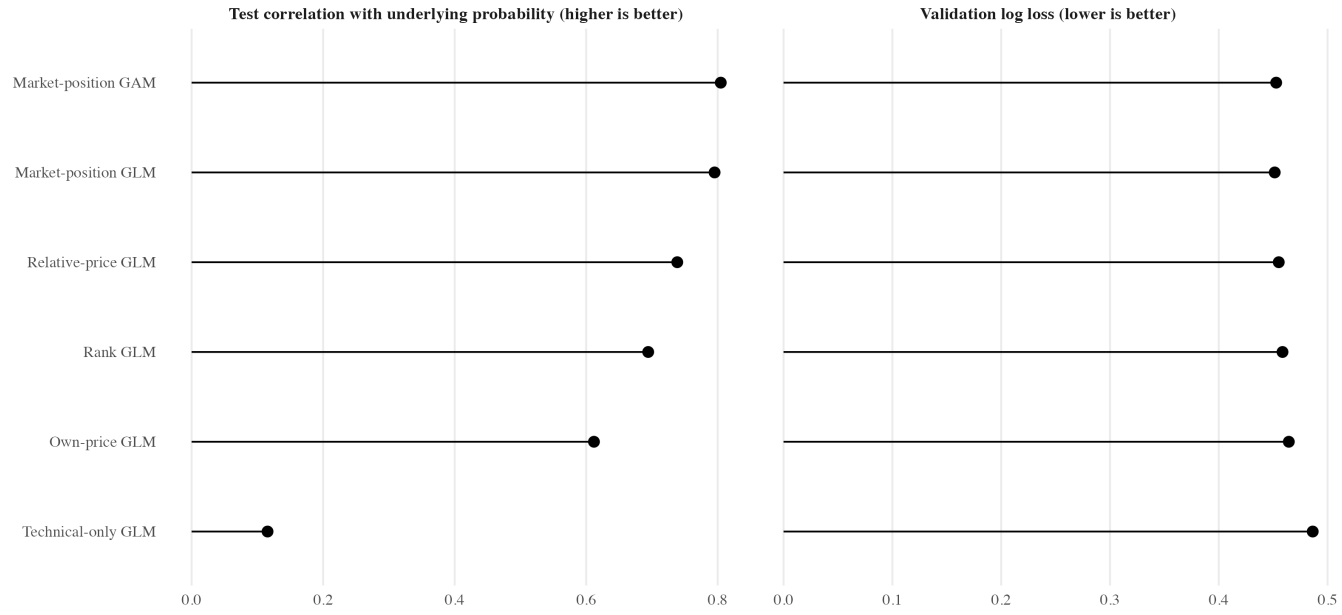


Figure 4: Incremental value of competitive information and nonlinear model flexibility.

4.3 Pricing Results

4.3.1 Selection of the Targeted Pricing Structure

Fifteen predefined segmentation variables were evaluated using the validation sample to identify a pricing structure that maximised expected conversion while satisfying the constraints described in Section 3.6.

Strategies were compared using two principal measures. **Relative conversion uplift** measures the percentage increase in expected conversion from the technical-pricing benchmark. **Contribution retention** compares expected contribution after claims under the proposed strategy with that under the benchmark. A retention value of 100% indicates no change, while values above 100% indicate higher expected contribution.

Both measures were calculated using fitted conversion probabilities and estimated claims costs during strategy selection.

Table 8 presents the five strongest admissible targeting structures.

Table 8: Leading admissible targeting structures on the validation sample

Targeting variable	Relative conversion uplift	Contribution retention
Vehicle-value band	19.79%	102.28%
Vehicle-age band	16.92%	103.05%
Vehicle segment	15.48%	100.85%
Annual-kilometres band	11.94%	101.93%
Unladen-weight band	11.66%	103.25%

Note: Results are based on fitted conversion probabilities and expected claims costs on the validation sample, relative to the technical-pricing benchmark.

Vehicle-value band achieved the largest admissible conversion improvement, with relative uplift of 19.79% and contribution retention of 102.28%. Although vehicle-age band produced higher contribution retention, vehicle-value band was selected because the optimisation prioritised expected conversion once the financial requirements were satisfied.

The selected strategy divided customers into four vehicle-value quartiles, with respective technical-premium multipliers of 0.95, 0.99, 1.00 and 1.01.

Consequently, the lowest-value quartile received a 5% reduction relative to technical premium, the second quartile received a 1% reduction, the third remained unchanged and the highest-value quartile received a 1% increase.

On the validation sample, the selected strategy produced an aggregate quoted rate level of 99.66%, representing a 0.34% reduction in total quoted premiums relative to the technical-pricing benchmark.

Expected contribution after claims increased by 2.28%, while the expected claims ratio was 77.67%, remaining below the 80% ceiling.

The lowest contribution retention among the four vehicle-value groups was 90.15%, narrowly exceeding the 90% minimum. This suggests that the minimum contribution requirement for each customer group restricted the extent to which expected contribution could deteriorate within particular customer groups.

The selection of vehicle-value band reflects its performance among the predefined segmentation variables under the simulated competitive and demand conditions. It does not demonstrate that vehicle value independently causes differences in customer price sensitivity.

The segmentation structure and its premium multipliers were fixed before evaluation on the independent test sample.

4.3.2 Independent Test Evaluation

The selected vehicle-value segmentation and premium multipliers were applied to the independent test sample. Performance was evaluated using the underlying simulated customer-choice probabilities, which had not been used to fit the conversion models or select the pricing strategy.

Under the technical-pricing benchmark, expected conversion was 18.28%. Targeted repricing increased this to 21.91%, representing an **absolute improvement of 3.63 percentage points** and a **relative conversion uplift of 19.88%**.

Across the 2,000 test profiles, this corresponded to approximately 73 additional expected conversions.

Table 9 compares the selected targeted strategy with the technical-pricing benchmark.

Table 9: Independent test sample comparison of technical and targeted pricing

Measure	Technical benchmark	Targeted strategy
Aggregate quoted rate level	100.00%	99.60%
Underlying expected conversion	18.28%	21.91%
Relative conversion uplift	—	19.88%
Expected contribution after claims	$82,378	$84,723
Contribution retention	100.00%	102.85%
Expected claims ratio	76.92%	77.73%

The strategy increased expected conversion while keeping total quoted premiums close to the technical-pricing benchmark. The aggregate quoted rate level of 99.60% represented a reduction of only 0.40%, indicating that the improvement was achieved primarily by redistributing premiums between customer groups rather than applying widespread discounts.

Expected contribution after claims increased from $82,378 to $84,723, representing an improvement of $2,345 and contribution retention of 102.85%.

These calculations use underlying simulated conversion probabilities and fitted expected claims costs. They therefore represent expected premium revenue less expected claims expenditure, rather than insurer profit after separately accounting for expenses and capital costs.

The expected claims ratio increased from 76.92% to 77.73% but remained below the specified 80% ceiling. This indicates that expected claims expenditure represented a slightly greater proportion of expected written premium following targeted repricing.

Nevertheless, the increase in expected conversion generated sufficient additional business to increase total expected contribution after claims. The strategy therefore improved expected conversion and total financial contribution despite a modest deterioration in the expected claims ratio.

For comparison, both uniform repricing strategies selected a 3% premium reduction across the portfolio, resulting in larger expected conversion improvements. However, their aggregate quoted rate level fell to approximately 97%, compared with 99.60% under the targeted strategy.

Because the uniform strategies used different optimisation objectives and were permitted to reduce aggregate quoted premiums more substantially, their outcomes are not directly comparable with the rate-constrained targeted strategy.

The targeted strategy therefore provides the principal evidence that, within the simulated market, reallocating premiums between customer groups can improve expected conversion while maintaining the specified aggregate pricing and financial requirements.

4.4 Robustness and Uncertainty

4.4.1 Structural Sensitivity

Structural sensitivity analysis examined whether the principal findings remained consistent when assumptions about competitive conditions, customer demand and pricing constraints were altered.

Ten predefined cases were evaluated, including the baseline.

Competitive market-position information improved conversion prediction and recovery of the underlying customer-choice probabilities relative to the Own-price GLM in all ten cases. However, the combined objective of increasing expected conversion while retaining at least 100% of benchmark expected contribution after claims was achieved in nine cases.

As illustrated in Figure 5, relative conversion uplift ranged from 1.19% under low customer price sensitivity to 32.60% under stronger price-threshold effects.

Contribution retention ranged from 99.50% to 111.43%, demonstrating that the financial benefits of targeted repricing depended on the assumed customer response to changes in premiums.

The low-price-sensitivity scenario was the only case that failed the combined pricing criterion when evaluated against the underlying customer-choice mechanism. Although competitive information continued to improve predictive performance, targeted repricing increased expected conversion by only 1.19%, while contribution retention declined to 99.50%.

This result is important because it demonstrates that improved conversion predictions do not necessarily translate into financially acceptable pricing decisions. When customers respond weakly to price changes, premium reductions may generate insufficient additional business to offset the corresponding reduction in contribution per accepted policy.

The remaining scenarios produced positive conversion improvements while retaining at least the benchmark expected contribution after claims.

Vehicle-value band remained the preferred targeting variable in eight of the ten cases, while the Market-position GAM was selected in eight cases and the Market-position GLM in two.

Overall, the predictive advantage of competitive information persisted across the examined assumptions. However, the size and financial acceptability of the resulting pricing improvements were less consistent, particularly when customer price sensitivity was reduced.

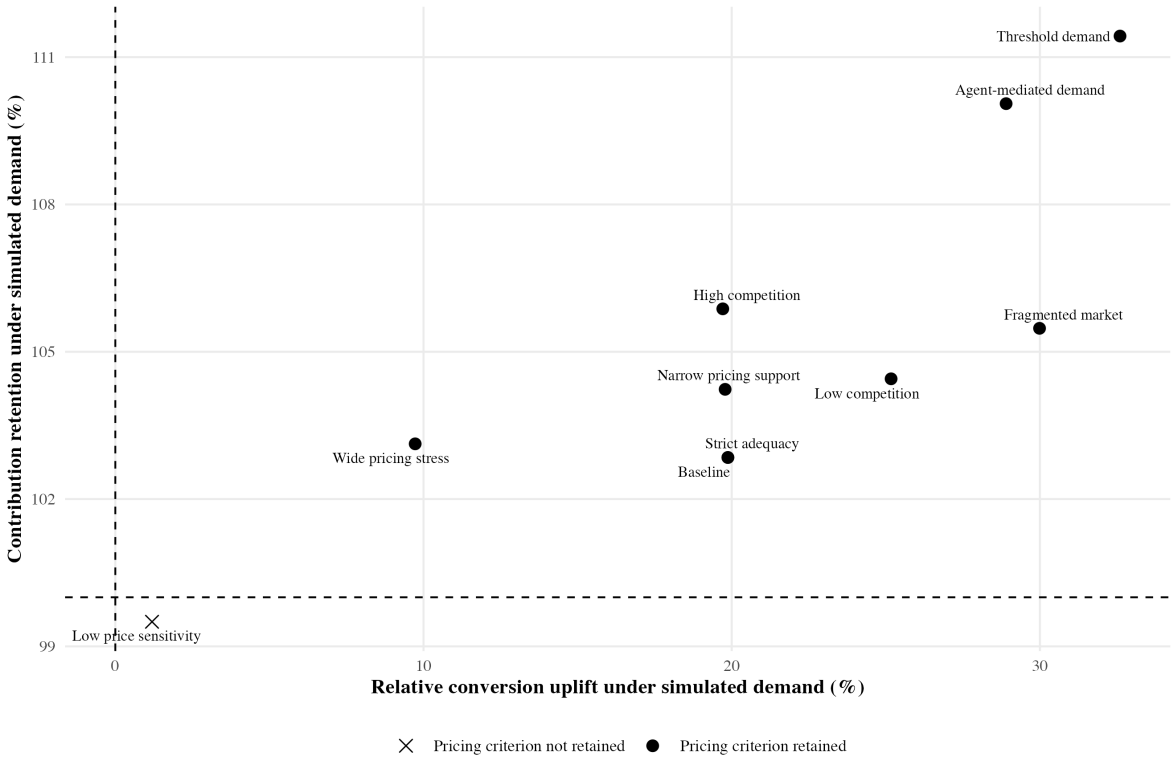


Figure 5: Structural sensitivity of expected conversion uplift and contribution retention under alternative market, demand and pricing assumptions.

4.4.2 Monte Carlo Regeneration

The complete simulation and modelling process was independently regenerated ten times to assess whether the results depended on the particular random observations in the baseline experiment.

Empirical calibration inputs and structural assumptions remained fixed, while new portfolios, claims experience, competing quotations and conversion outcomes were generated for each replication.

The selected Market-position model outperformed the Own-price GLM in all ten replications. Improvements in independent test log loss ranged from 1.71% to 3.43%, averaging 2.61%. Recovery of underlying conversion probabilities also improved consistently.

These findings indicate that the predictive value of competitive information was not specific to the initial synthetic dataset.

Targeted repricing produced positive expected conversion improvements in every replication. Relative conversion uplift ranged from 15.57% to 25.48%, averaging 22.83%, while contribution retention ranged from 102.44% to 104.23%, with a mean of 103.29%.

Vehicle-value band was selected as the preferred targeting variable in all ten replications, indicating that the baseline targeting choice was reasonably stable when the simulation was repeated under unchanged structural assumptions.

However, the resulting test aggregate quoted rate levels ranged from 99.38% to 99.61%, with three replications falling slightly below the 99.5% lower bound applied during validation-based optimisation.

This does not necessarily indicate that the selected strategies violated the constraints during optimisation. The premium adjustments were selected using the validation samples, whereas the reported rate levels were calculated on independently generated test samples. Differences in the characteristics of customers across these samples can therefore cause test-sample aggregates to fall outside the validation constraints.

The preferred conversion-model family was less stable. Although the Market-position GAM was selected in the baseline experiment, the Market-position GLM was selected in nine of the ten Monte Carlo replications.

The GAM's average validation log-loss improvement over the GLM was only 0.21%, indicating that the incremental benefit of nonlinear flexibility was sensitive to random variation.

These results reinforce the distinction between the value of competitive market information and increased model complexity. Competitive information consistently improved predictive performance and supported favourable targeted-pricing outcomes, while the preference for the more flexible GAM was not consistently reproduced.

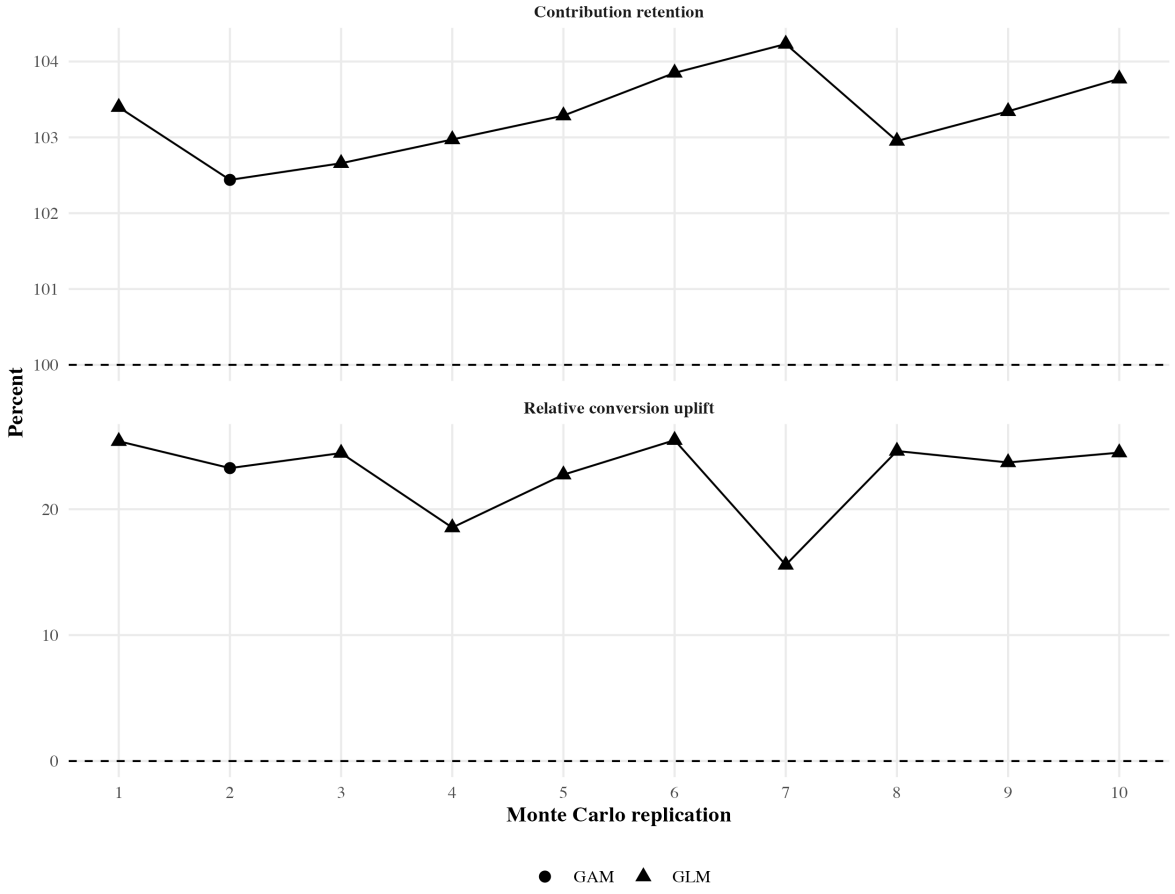


Figure 6: Stability of expected conversion uplift and contribution retention across ten independent Monte Carlo replications.

4.4.3 Paired Bootstrap

A paired bootstrap was conducted to assess whether the final pricing results depended on the particular 2,000 customers included in the independent test sample.

The analysis used 5,000 resamples drawn with replacement, retaining each customer's technical-pricing benchmark and targeted-pricing outcomes as a pair. The selected pricing strategy, fitted models and claims-cost estimates remained unchanged throughout. Table 10 presents the resulting 95% bootstrap percentile intervals.

Table 10: Paired-bootstrap uncertainty for the independent test sample pricing comparison

Measure	Point estimate	95% bootstrap interval
Absolute conversion uplift	3.63 pp	3.34–3.94 pp
Relative conversion uplift	19.88%	18.29–21.46%
Contribution retention	102.85%	102.09–103.64%
Targeted claims ratio	77.73%	77.65–77.82%
Aggregate quoted rate level	99.60%	99.52–99.68%

Note: Intervals are based on 5,000 paired resamples of the independent test sample. The selected pricing strategy and fitted claims-cost estimates remain fixed. The intervals therefore describe variation associated with the sampled customers rather than uncertainty in the underlying customer-choice assumptions.

The estimated absolute conversion improvement was 3.63 percentage points, with a 95% bootstrap interval of 3.34–3.94 percentage points. The interval remained entirely above zero, indicating that the estimated improvement was not strongly dependent on the particular customers included in the original test sample.

Contribution retention ranged from 102.09% to 103.64%, remaining above the 100% benchmark throughout the interval.

All 5,000 bootstrap resamples produced positive conversion improvements and retained at least the benchmark expected contribution after claims.

The expected claims ratio remained below the 80% constraint, with an interval of 77.65%–77.82%. The aggregate quoted rate level ranged from 99.52% to 99.68%, also remaining within the specified bounds, although its lower endpoint was close to the minimum of 99.5%.

These results provide additional evidence that the favourable conversion and contribution outcomes were not driven by a small number of customer profiles in the test sample.

However, the bootstrap holds the pricing decisions and simulated customer-choice relationships fixed. The intervals therefore do not capture uncertainty about the underlying demand model, alternative pricing strategies or outcomes in an observed insurance market.

Taken together, the robustness analyses indicate that competitive information consistently improved conversion modelling across the examined scenarios and independently generated datasets. Targeted repricing also generally improved expected conversion while preserving expected contribution after claims, although its financial effectiveness depended on the assumed strength of customer price responses.

5 Discussion

The results demonstrate that competitive market-position information can improve conversion modelling and support more effective pricing decisions within the synthetic insurance environment. The most consistent improvements arose from incorporating information about competing quotations, rather than increasing model complexity. However, the pricing results also demonstrate that improved predictive performance does not necessarily produce financially beneficial premium adjustments. This section considers these findings, their practical implications and the limitations of the synthetic framework.

5.1 Competitive Information and Model Complexity

A central finding is that incorporating competitive market-position information provided greater and more consistent improvements in conversion prediction than introducing additional nonlinear modelling flexibility. The Own-price GLM outperformed the Technical-only GLM, demonstrating the value of incorporating the reference insurer's quoted premium alongside customer characteristics and technical risk. Further improvements occurred when relative price, quotation rank and other competitive-position measures were included.

This finding is consistent with the distinction between technical and commercial insurance pricing discussed by Taylor (1986), Guelman and Guillén (2014) and Spedicato et al. (2018). Technical pricing estimates the expected cost of insuring a customer, whereas commercial pricing must also account for the likelihood of purchase. In a comparison market, that likelihood may depend not only on the premium offered but also on the alternatives available. Consequently, the same premium can represent a favourable or unfavourable offer depending on competing quotations. Competitive-position variables provide information about this relationship that cannot be obtained from the reference insurer's premium alone.

The comparison between the Market-position GLM and GAM provides a separate insight into model complexity. Although the GAM achieved the strongest validation performance in the baseline experiment, its 0.51% improvement in log loss over the corresponding GLM was relatively small. Furthermore, the Market-position GLM was selected in nine of the ten Monte Carlo replications, indicating that the additional benefit of nonlinear flexibility was not consistently reproduced.

By contrast, incorporating competitive information improved predictive performance across every structural sensitivity case and Monte Carlo replication. This suggests that, within the simulated environment, providing a model with more relevant information was generally more valuable than increasing its flexibility. Where a simpler model captures most of the relevant relationships, the additional complexity of a GAM may offer limited improvement relative to the greater demands it places on validation, interpretation and governance.

Nevertheless, these results do not establish that GLMs are generally preferable to GAMs. Their relative performance depends on the relationships governing customer behaviour and the information available for estimation. In particular, competitive effects were explicitly incorporated into the simulation. The findings therefore demonstrate that the models could recover these assumed relationships, rather than establishing their prevalence or magnitude in observed Australian motor insurance markets.

5.2 From Predictive Performance to Pricing Decisions

Improved conversion predictions are valuable for pricing only when they support better premium decisions. A model may achieve lower log loss or stronger discrimination without accurately estimating how customers would respond to alternative prices. Consequently, predictive accuracy alone is insufficient to establish whether a conversion model can improve pricing outcomes.

This distinction is particularly important in competitive markets because changing a premium also changes its position relative to other offers. For example, a modest premium reduction may move the reference insurer from the second-cheapest to the cheapest quotation, potentially producing a larger response than an equivalent reduction that leaves its competitive rank unchanged. Evaluating alternative premiums therefore requires competitive-position variables to be recalculated rather than held fixed.

The pricing framework addressed this by updating relative price, rank and other competitive measures for each proposed premium. The fitted conversion models were used to select pricing adjustments, while the resulting decisions were evaluated on an independent test sample using the underlying simulated customer-choice probabilities. This separation is important because favourable predictions from a fitted model do not necessarily imply favourable outcomes under the process that actually generates customer behaviour.

The targeted strategy increased expected conversion by 19.88% and expected contribution after claims by 2.85%. These improvements remained when assessed using the underlying customer-choice mechanism, rather than relying solely on the fitted model's predicted probabilities. The result therefore provides evidence that the selected strategy was effective under the simulation's specified demand relationships.

This ability to evaluate alternative prices is a methodological advantage of the synthetic framework. In observed insurance data, purchasing outcomes are generally available only for premiums actually offered, making it difficult to determine how the same customer would have responded to another price. The simulation provides the underlying probabilities needed to make these comparisons directly. Its value therefore lies in supporting controlled evaluation of pricing methods, consistent with the emphasis on assessing synthetic data according to their intended analytical purpose (Jordon et al., 2024; Jones et al., 2025).

However, this does not resolve uncertainty about customer behaviour in an observed market. The evaluation establishes that the pricing strategy improved expected outcomes under the assumed customer-choice mechanism, not that the same changes would occur among Australian insurance customers.

5.3 Interpretation of Targeted Repricing

The pricing results must be interpreted alongside the objectives and constraints imposed during optimisation. Reducing premiums can improve the reference insurer's competitive position and encourage additional purchases, but it also reduces the contribution earned on each policy sold, unless the increase in demand is sufficient to offset the lower premium.

The uniform-pricing strategies illustrated this trade-off by producing larger conversion improvements through a 3% reduction in premiums across the portfolio. However, these strategies were optimised using a different objective and were not subject to the narrow aggregate quoted rate-level restriction applied to targeted repricing. Their results therefore provide a useful comparison, but not a directly equivalent alternative under the same constraints.

Targeted repricing instead sought to increase expected conversion while keeping total quoted premiums close to the technical-pricing benchmark and satisfying the specified financial requirements. The selected strategy achieved this by applying different multipliers across vehicle-value quartiles rather than reducing premiums uniformly.

Vehicle-value band was selected because it produced the largest admissible conversion improvement among the fifteen predefined segmentation variables. Its selection reflects the combined effects of expected claims costs, competitive positioning and predicted customer responses within the simulated market. It should not be interpreted as evidence that vehicle value independently determines customer price sensitivity.

The results for the individual vehicle-value groups illustrate how these factors interact. On the validation sample, the lowest vehicle-value quartile retained 133.34% of its benchmark expected contribution after claims despite receiving the largest premium reduction of 5%. Conversely, the highest vehicle-value quartile retained approximately 90.15% despite receiving a 1% premium increase.

These outcomes arise because expected contribution after claims depends on both the difference between the premium and expected claims cost and the probability that the customer purchases the policy. A lower premium may increase expected contribution if it attracts sufficient additional business. Similarly, a higher premium may reduce expected contribution if the resulting decline in conversion outweighs the additional amount charged per accepted policy.

The overall pricing results reinforce this distinction. Following targeted repricing, the expected claims ratio increased from 76.92% to 77.73%, indicating that expected claims expenditure represented a slightly greater proportion of expected written premium. Nevertheless, the increase in expected conversion generated sufficient additional business to raise total expected contribution after claims.

This demonstrates why pricing strategies should be assessed using several complementary measures. Conversion captures the expected volume of business, the expected claims ratio assesses claims adequacy relative to written premium, and expected contribution after claims measures the combined financial effect of premium amounts and expected customer purchases. An improvement in one measure does not necessarily imply an improvement in the others.

Vehicle-value segmentation also remained reasonably stable across the robustness analyses, although alternative variables were preferred under some structural assumptions. This suggests that the benefit of reallocating premiums between customer groups may be more robust than the choice of a particular segmentation characteristic. In an operational setting, the suitability of the selected segments would require continued evaluation as customer behaviour, claims experience and competitive conditions change.

5.4 Robustness and Boundary Conditions

The robustness analyses provide additional evidence that the principal findings were not dependent on a single set of simulated observations or assumptions. However, they also identify conditions under which improved conversion modelling does not translate into financially acceptable pricing outcomes.

The most consistent finding was the predictive value of competitive information. Market-position models outperformed the Own-price GLM across all ten structural sensitivity cases and all ten independent Monte Carlo replications. This indicates that the improvement was not confined to the original synthetic dataset or the baseline specification of customer demand.

The financially benefits of targeted repricing were less consistent. The combined objective of increasing expected conversion while retaining at least the benchmark expected contribution after claims was achieved in nine of the ten structural sensitivity cases.

The principal exception was the low-price-sensitivity scenario, where relative conversion uplift fell to 1.19% and contribution retention declined to 99.50%. Although competitive information continued to improve conversion predictions, the additional expected purchases were insufficient to preserve benchmark contribution.

This distinction is important. A model may accurately identify how competitive position affects customer demand while providing limited opportunities for financial beneficial repricing. When customers respond weakly to price changes, reductions in premium may generate insufficient additional purchases to compensate for the lower contribution per policy. The value of competitive information for prediction therefore does not guarantee that changing premiums will improve financial outcomes.

The Monte Carlo results further demonstrated that favourable pricing outcomes were not specific to a single generated portfolio. Targeted repricing increased expected conversion and contribution after claims in all ten replications, while vehicle-value band was consistently selected as the preferred targeting variable.

Nevertheless, three replications produced aggregate quoted rate levels on the independent test samples slightly below the 99.5% minimum applied during optimisation. This reflects the distinction between satisfying constraints on the validation sample and obtaining the same aggregate measures on a separate group of customers. It demonstrates that financial requirements satisfied during strategy selection may not remain within identical bounds when the selected prices are applied to a different portfolio composition.

The paired bootstrap provided further evidence of stability. The 95% interval for absolute conversion uplift remained positive, while the interval for contribution retention remained above 100%. These results indicate that the baseline improvements were not strongly dependent on the particular customer profiles included in the test sample.

However, the bootstrap held the fitted models, selected pricing strategy and underlying customer-choice relationships fixed. It therefore measures uncertainty associated with the composition of the test sample rather than uncertainty in model selection or the assumptions used to simulate demand.

Taken together, the robustness analyses provide stronger evidence for the predictive value of competitive information than for the financial benefits of any particular repricing strategy. The latter depend more substantially on customer price responsiveness and the financial constraints imposed during optimisation.

5.5 Practical Implications

The principal practical implication is that conversion modelling should complement, rather than replace, technical pricing. Expected claims costs provide the actuarial basis for assessing premium adequacy, while conversion models estimate the likelihood that customers will accept the premiums offered. Considering both components allows pricing decisions to account for customer acquisition and expected financial contribution.

For insurers operating in comparison-based markets, competitor quotations may provide useful information beyond that contained in their own prices. Measures such as relative price and quotation rank describe the alternatives available to individual customers and may help explain differences in purchasing behaviour. However, obtaining these benefits in practice would depend on the availability, reliability and representativeness of competitive quotation data.

The modelling results also suggest that improving the information available to a conversion model may be as important as adopting more sophisticated statistical techniques. In the simulated environment, a Market-position GLM captured most of the predictive improvement achieved by the GAM, while offering a simpler modelling structure. This supports beginning with interpretable approaches and introducing greater flexibility when it produces meaningful improvements in predictive performance or pricing outcomes.

The optimisation results further emphasise that customer acquisition should not be evaluated in isolation. Premium reductions may attract additional business while weakening claims adequacy or reducing contribution within particular customer groups. Practical pricing decisions would therefore require explicit financial requirements and monitoring financial results across all customers and within individual customer groups.

Governance and customer fairness are also important considerations. A statistically effective segmentation characteristic is not necessarily appropriate for insurance pricing, particularly if its use produces unjustified differences between customer groups. Any operational application would require assessment of regulatory obligations, explainability, discrimination risks and customer outcomes, consistent with the considerations discussed by the Australian Human Rights Commission and Actuaries Institute (2022).

Finally, the study demonstrates how synthetic experimentation can support the development of pricing methods when proprietary competitor quotations and conversion outcomes are unavailable. The framework allows researchers to test modelling choices, optimisation constraints and evaluation procedures before applying them to observed insurance data.

Its practical value is therefore methodological: it provides a controlled and reproducible approach for investigating demand-aware pricing, rather than premium adjustments that insurers could adopt directly.

5.6 Limitations

The principal limitation is that customer purchasing behaviour is simulated rather than observed. The underlying customer-choice mechanism explicitly incorporates responses to relative price, quotation ranking and competitive thresholds. Consequently, the improvements achieved by Market-position models demonstrate their ability to recover these assumed relationships, but do not establish that Australian motor insurance customers exhibit the same responses.

The reported conversion improvements should therefore not be interpreted as empirical estimates of the benefits available to insurers operating in the Australian market.

The synthetic competitive market is also simplified. Competing insurers are represented through specified pricing strategies based on a common technical-premium reference, rather than independently estimated claims experience and pricing models. Actual insurers may differ in their underwriting objectives, expense structures, target returns, claims expectations and capital requirements.

Furthermore, competitor quotations remain unchanged when the reference insurer adjusts its premiums. The framework therefore does not capture strategic responses by competing insurers, which could influence the effectiveness of pricing adjustments over time.

Although Australian public data inform the synthetic portfolio, the generated policies do not represent an observed insurer's portfolio. Some demographic, vehicle and geographic characteristics are calibrated to published statistics, while other underwriting relationships and risk adjustments rely on modelling assumptions. The technical-pricing results consequently demonstrate the recovery of expected claims costs within the constructed environment, rather than establishing the accuracy of the assumed risk relationships in an operational insurance market.

The scope of the pricing experiment introduces further limitations. Premium adjustments are restricted to relatively small changes around technical pricing, and the underlying customer-choice mechanism is assumed to remain stable when these changes are applied. The findings therefore provide limited evidence about larger premium adjustments or markets in which customer behaviour changes substantially over time.

The analysis also focuses on new-business conversion rather than renewal retention. Existing policyholders may respond differently to premium changes because of switching costs, familiarity with their insurer and other behavioural factors. Similarly, the customer-choice model represents differences in premiums and stylised insurer preferences without explicitly modelling variation in coverage, service quality, reputation or optional benefits.

The financial evaluation is restricted to expected contribution after claims. Although the technical premium includes a general 30% loading, the optimisation does not separately model operating expenses, acquisition costs, reinsurance, capital charges, taxation or investment income. The reported contribution improvements therefore do not represent changes in net insurer profitability or establish the solvency implications of the selected strategy.

Finally, the robustness analyses cannot eliminate uncertainty arising from unexamined market conditions. Structural sensitivity evaluates a predefined set of alternative assumptions, but does not comprehensively examine their interactions. The ten Monte Carlo replications provide a useful assessment of stability across independently generated datasets, although they are insufficient for precise estimation of simulation uncertainty. The paired bootstrap evaluates variation across customer profiles while holding model selection, pricing decisions and the underlying demand relationships fixed.

These procedures strengthen confidence in the internal reliability of the findings but do not establish their external validity. Applying the framework to an operational insurance market would require observed quotation and conversion data to validate customer-demand relationships, empirical assessment of competitor pricing and a more comprehensive financial and governance evaluation.

6 Conclusion

This dissertation investigated whether a fully synthetic Australian motor insurance framework could represent a competitive quotation market and whether incorporating competitive market-position information could improve conversion modelling and pricing decisions. The findings provide support for both research objectives within the assumptions of the simulated environment.

The first research question was addressed by developing a synthetic portfolio that integrated policyholder characteristics, claims experience, technical premiums, competing quotations and customer purchasing decisions. The technical-pricing models recovered substantial differences in underlying expected claims costs, while the quotation market generated consistent relationships between premiums, competitive position and conversion. This demonstrated that a coherent environment for examining competitive insurance pricing could be constructed without access to proprietary quotation and conversion data. However, its consistency with the specified simulation assumptions does not establish that the generated customer behaviour accurately represents the Australian motor insurance market.

The second research question examined whether competitive information improved conversion predictions and supported more effective pricing decisions. Incorporating the reference insurer's own premium improved predictive performance relative to technical-risk characteristics alone, while adding competitor quotations and market-position variables produced further improvements. The selected Market-position GAM achieved a Spearman correlation of 0.817 with underlying conversion probabilities on the independent test sample, compared with 0.623 for the Own-price GLM. These findings demonstrate the additional predictive value of competitive information within the simulated market.

The pricing results showed that these improvements could translate into more effective premium decisions. Targeted repricing increased expected conversion from 18.28% to 21.91%, representing an absolute increase of 3.63 percentage points and a relative improvement of 19.88%. The strategy maintained total quoted premiums close to the technical-pricing benchmark while increasing expected contribution after claims by 2.85%. The expected claims ratio remained below the specified 80% ceiling. Importantly, these outcomes were evaluated using the underlying simulated customer-choice probabilities, rather than relying solely on the predictions used to select the premium adjustments.

The robustness analyses provided further insight into the reliability and limitations of these findings. Competitive market-position information improved predictive performance across all structural sensitivity cases and Monte Carlo replications. By contrast, the incremental benefit of greater model complexity was less consistent: although the Market-position GAM was selected in the baseline analysis, the Market-position GLM was preferred in nine of the ten Monte Carlo replications. The financial benefits of targeted repricing were also sensitive to the assumed customer response to premium changes, with the low-price-sensitivity scenario failing to preserve benchmark expected contribution after claims. These findings indicate that improved conversion modelling does not necessarily guarantee financially beneficial pricing decisions.

The conclusions remain conditional on the synthetic framework. Customer demand and competitor pricing were generated using specified assumptions rather than estimated from observed market behaviour, and the financial evaluation excluded separately modelled expenses and capital costs. Accordingly, the reported conversion improvements and selected premium adjustments should not be interpreted as estimates of outcomes achievable by Australian motor insurers.

The principal contribution of this research is therefore methodological. It establishes a reproducible framework that combines technical pricing, competitive quotation information, conversion modelling and constrained premium optimisation. By separating the relationships used to generate synthetic outcomes from the information available during model development and pricing selection, the framework also enables pricing decisions to be evaluated independently against the simulated customer-choice mechanism. This provides a foundation for investigating demand-aware insurance pricing methods when complete competitive quotation and conversion data are unavailable.

6.1 Further Work

The most important extension would be to calibrate and validate the framework using observed competitor quotations and customer conversion outcomes. Such data would allow the assumed relationships between premiums, competitive position and purchasing behaviour to be assessed against actual market experience. This would be necessary to determine whether the predictive improvements and pricing benefits identified in the synthetic experiment extend to observed Australian motor insurance markets.

Future research could also develop a more detailed representation of insurance competition. Rather than generating competing quotations from a common technical-premium reference, each competitor could be assigned an independent risk model, pricing strategy and financial objective. Allowing insurers to respond to one another's premium changes would provide a more realistic representation of strategic competition. Incorporating differences in coverage, service quality and insurer reputation could further improve the representation of customer purchasing decisions.

An additional extension would be to examine renewal business alongside new-business conversion. Existing policyholders may respond differently to premium changes because of customer loyalty, switching costs and previous experience

with their insurer. Modelling these effects would allow the framework to investigate pricing decisions across a broader range of insurance customer relationships.

Finally, the robustness and financial evaluation could be extended. Multi-factor sensitivity analysis would allow interactions between competitive conditions, customer demand and pricing constraints to be examined, while additional Monte Carlo replications could provide more precise estimates of simulation uncertainty. Incorporating acquisition and operating expenses, reinsurance, capital requirements and broader profitability measures would also support a more comprehensive evaluation of pricing decisions. These developments, together with assessment of regulatory requirements and customer fairness, would improve the framework's suitability for practical insurance pricing applications.

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A Document Conventions

The tables below define the principal abbreviations and mathematical symbols used in the dissertation. They distinguish the underlying relationships used to generate synthetic claims and customer decisions from the fitted models used to estimate expected claims costs, predict conversion and select premiums.

A.1 Acronyms and Abbreviations

Table A.1: Acronyms and abbreviations

Abbreviation	Definition
ABS	Australian Bureau of Statistics
AUC	Area under the receiver operating characteristic curve
BITRE	Bureau of Infrastructure and Transport Research Economics
CV	Coefficient of variation
DGP	Data-generating process
FCAI	Federal Chamber of Automotive Industries
GAM	Generalised additive model
GLM	Generalised linear model
MAE	Mean absolute error
MC	Monte Carlo
O/E	Observed-to-expected ratio
RMSE	Root mean squared error

Two distinct data splits are used. Technical-pricing candidates are fitted on the 2021–2023 cohorts, selected on the 2024 cohort, refitted on 2021–2024, and evaluated on the 2025 cohort. The 2025 quotation cohort is separately divided into a **training sample** (6,000 profiles), **validation sample** (2,000 profiles) and **test sample** (2,000 profiles) for conversion modelling and pricing evaluation. The terms training, validation and test refer to this quotation-level split unless the technical-pricing cohort is specifically identified.

A.2 Summary of Notation

Table A.2: Principal claims and technical-pricing notation

Symbol	Definition
i	Policy or quotation-profile index
j	Reported-claim index
N_i	Number of **reported** claims for policy i in the synthetic claims DGP
C_i	Number of **paid claims** for policy i: reported claims with insurer payment greater than zero
λ_i	Underlying annual **reported-claim** frequency
R_i^A, R_i^L, R_i^K	Driver-age, licence-experience and annual-kilometres frequency relativities
$G_{\ell(i)}$	Shared geographic frequency effect for synthetic locality $\ell(i)$
U_i	Policy-specific Gamma frequency frailty
T_{ij}	Indicator that reported claim j is a **write-off**
X_{ij}	Gross claim cost before deducting voluntary excess
e_i	Policyholder voluntary excess
$Y_{ij} = \max(X_{ij} - e_i, 0)$	Insurer payment for a reported claim, possibly zero
B_i	Reference new-vehicle price on the 2025 price basis

Symbol	Definition
V_i, V_0	Simulated depreciated vehicle value and reference vehicle value on the 2025 price basis
s_i	Underlying expected insurer payment **per reported claim**, including zero-payment claims
$\mu_i = \lambda_i s_i$	Underlying expected annual insurer claims cost
$\widehat{\lambda}_i, \widehat{s}_i$	Fitted annual paid-claim frequency and mean insurer payment per paid claim
$\widehat{\mu}_i = \widehat{\lambda}_i \widehat{s}_i$	Fitted expected annual claims cost
$\pi_i = 1.30\widehat{\mu}_i$	Fitted technical premium, including the 30% loading
E_i	Policy exposure; one year per simulated profile
A_i, L_i, K_i	Driver age, licence experience in years and annual kilometres

Table A.3: Principal quote, demand and optimisation notation

Symbol	Definition
k, h	Competitor index and insurer-alternative index, respectively
P_i	Initial reference insurer quotation; also used for a proposed premium when the pricing context is explicit
Q_{ik}	Quotation returned by competitor k
$Q_i^{\min}$	Cheapest **competing** quotation returned for profile i
ω_i	Initial reference insurer commercial-price multiplier, Uniform$(0.95, 1.05)$
$\alpha_k, \delta_{ik}, \varepsilon_{ik}$	Competitor log-price stance, segment adjustment and stochastic quote variation
$\mathcal{K}_i$	Set of competitors returning a quote to customer i
$\mathcal{J}_i$	Complete available insurer choice set, including the reference insurer
$RP_i = P_i/Q_i^{\min}$	Initial reference insurer premium relative to the cheapest competitor
R_i	Initial reference insurer quote rank; one denotes the cheapest quotation
q_i	Underlying probability of purchasing from any available insurer in the comparison market
w_{ih}	Conditional probability of choosing insurer alternative h after deciding to purchase
w_i	Conditional probability of choosing the reference insurer, given a market purchase
U_{ih}	Customer-choice utility for alternative h
β_i	Customer-specific price sensitivity in the choice DGP
$p_i = q_i w_i$	Underlying reference insurer conversion probability at the initial quotation
D_i	Realised binary reference insurer conversion outcome
$\widehat{p}_i(P_i)$	Fitted conversion probability at proposed reference insurer premium P_i
$p_i(s)$	Underlying reference insurer conversion probability under pricing strategy s
m_i	Multiplier applied to fitted technical premium during repricing
x_{gm}	Binary indicator that price multiplier m is selected for segment g
RL	Aggregate quoted rate level: total proposed premiums divided by total technical premiums

Symbol	Definition
$\widehat{M}$	Model-expected contribution after claims, $\sum_i \hat{p}_i(P_i)(P_i - \hat{\mu}_i)$
M_s	Expected contribution after claims under strategy s, using underlying simulated conversion probabilities and fitted claims costs
CR	Expected claims expenditure divided by expected written premium, both weighted by conversion probability

Claims definitions: A *reported claim* includes claims whose gross cost is fully absorbed by the policyholder's excess. A *paid claim* results in a strictly positive insurer payment. The simulation uses reported-claim frequency λ_i and expected payment per reported claim s_i, while the fitted technical-pricing model uses paid-claim frequency $\widehat{\lambda}_i$ and severity $\hat{s}_i$. Both approaches estimate annual insurer claims expenditure.

Financial definitions: Expected contribution after claims is conversion-probability-weighted premium revenue less conversion-probability-weighted **fitted** expected claims costs. It excludes separately modelled operating and acquisition expenses, reinsurance costs and capital charges, and is **not** net insurer profit. The aggregate quoted rate level instead considers all offered premiums, including quotations that are not accepted. The cheapest competing quotation used for the reference insurer's relative price may differ from the cheapest quotation across the complete choice set used in insurer-choice utility.

B Calibration and Simulation Assumptions

The 50,000 profile synthetic portfolio comprises 10,000 independent one-year new-business policies for each calendar cohort from 2021 to 2025. Australian public data inform selected observable characteristics; unobserved claims heterogeneity, competitor behaviour and customer choice are governed by explicit assumptions. Empirical inputs are not evidence that the joint synthetic population reproduces the insured Australian market.

B.1 Empirical Inputs and Portfolio Calibration

Table B.1: Empirical calibration register

Component	Source or basis	Implementation and qualification
State and registered-vehicle composition	BITRE annual *Road Vehicles Australia* statistics	Annual State and vehicle manufacture-year/make/model calibration; probabilities vary by cohort where annual information is available.
Remoteness conditional on State	BITRE postcode vehicle counts; ABS ASGS Edition 3 allocation files and 2021 Census dwelling counts	Five remoteness categories; overlapping postal areas allocated with dwelling weights. The empirically derived **2025** $P(\text{Remoteness} \mid \text{State})$ is held constant across all five cohorts, while annual State shares vary.
Driver ages	BITRE *Australian Infrastructure and Transport Statistics Yearbook 2025*, licence-holder age bands	Drawn from the national licence-holder structure. **50%** of the source 15–19 band is retained to represent eligible 18–19-year-olds; this frozen approximation differs from the 40% implied by equal numbers at each single age.

Component	Source or basis	Implementation and qualification
Vehicle segment and motive power	Australian Government Green Vehicle Guide; FCAI classification criteria; BITRE registrations	Curated make/model/year classification and manufacture-year-conditional motive-power distribution.
Physical vehicle specifications	CarExpert reference records and compatibility-based donor matching	Engine capacity, cylinder count, power, torque, transmission and unladen weight. The code uses a lookup named `carexpert_vehicle_spec_lookup_95PCT.csv` and checks high coverage; that specific input must be included with the computational materials.
Vehicle value	RACQ *Vehicle Running Costs 2025*	Reference new prices and five-year value ratios; interpolated/extrapolated depreciation is a simulation assumption, with a $3,000 floor and new-price ceiling.
Vehicle usage	ABS (2021) *Electric passenger vehicle use, experimental estimates*	Mean annual distance calibrated to **11,100 km** based on Australian passenger vehicles for the 12 months ended 30 June 2020. Petrol (**10,500 km**), diesel (**13,900 km**) and battery-electric (**11,100 km**) evidence informs motive-power relativities. Hybrid and other-fuel relativities are neutral assumptions; the empirical battery-electric value is applied to the combined battery/fuel-cell category. Driver-age/remoteness effects and lognormal CV **0.45** are assumed.
Calendar-year monetary levels	ABS motor-vehicle price and vehicle-maintenance/repair indices	Write-off values and partial-claim costs respectively converted to the common 2025 pricing basis.
Underwriting characteristics	Explicit simulation distributions	Voluntary excess, parking, financing, modification, pre-existing damage and regular-driver status; not observed insurer prevalences.

Approximately 450 **synthetic localities** receive a shared random frequency effect. These are simulation groupings, not observed postcode-level risk relativities. Vehicle ages are capped at 30 years. The national annual-kilometre benchmark is taken from ABS vehicle-use estimates for 2019–20 (Australian Bureau of Statistics, 2021b) and applied to each synthetic cohort; it is not an estimate of average driving in every year from 2021 to 2025. Voluntary excess is sampled using the probabilities stated below.

Table B.2: Selected scalar generation and claims assumptions

Parameter	Baseline	Interpretation
Policy count / cohorts	50,000; 2021–2025	10,000 independent policies per year
Annual policy exposure	1 year	Applied to each profile
Synthetic geographic groups	About 450	Shared local frequency heterogeneity
Driver-age effect	$\min\{2.20, \exp[0.00055(A_i - 50)^2]\}$	Minimum at age 50; cap 2.20

Parameter	Baseline	Interpretation
Licence effect	$\exp[-0.020\min(L_i, 15)]$	Reduction capped at 15 years licensed
Kilometre effect	$(K_i/11{,}100)^{0.55}$	Diminishing frequency effect of annual kilometres
Geographic frequency effect	Lognormal mean 1; log-SD 0.15	Shared across each synthetic locality
Individual frequency frailty	Gamma mean 1; variance 0.20	Unobserved policy risk
2025 claimant target	14%	Probability of at least one **reported** claim
Write-off claim cost share	25%	Share of total **insurer-paid claim expenditure**, not share of claim counts
Average insurer payment per reported claim	Approximately \$5,202	**Includes zero-payment claims absorbed by excess**; 2025 basis
Partial-claim gross severity	Lognormal; CV 1.00	Conditional on a claim not being a write-off
Partial-claim vehicle-value exponent	0.35	Relativity $(V_i/V_0)^{0.35}$
Mean annual kilometres	Approximately 11,100	Based on the ABS 2019–20 passenger-vehicle average; imposed in each synthetic policy year, subject to rounding and bounds
Annual-kilometres CV	0.45	Initial lognormal simulation assumption
Vehicle-age cap / value floor	30 years / \$3,000	Synthetic portfolio/value constraints
Technical-premium loading	30%	$\pi_i = 1.30\hat{\mu}_i$; implies fitted-basis claims ratio 76.92%

Claims calibration evidence: The Insurance Council of Australia's *Motor Insurance Policy Paper* (2025, pp. 7, 12 and 17) reports approximately one in seven policyholders making a claim annually, an average claim size of \$5,202 as at June 2024, and write-offs accounting for approximately one-quarter of claims expenditure. These figures inform the simulation but do not determine its parameters uniquely. Module 01 applies the \$5,202 target on a **2025 price basis** to the average insurer payment per **reported claim**, including claims with zero insurer payment. The denominator and price basis do not necessarily match the published industry claim-size statistic. Accordingly, the study uses the industry figure as an indicative calibration target rather than claiming an exact like-for-like match.

Voluntary excess and claim calibration: Voluntary excesses are \$400 (10%), \$600 (25%), \$800 (35%), \$1,000 (20%) and \$1,500 (10%). The insurer payment for a claim is $Y_{ij} = \max(X_{ij} - e_i, 0)$. The write-off probability and partial-claim severity scale are calibrated jointly to the average-payment and write-off-expenditure-share targets after allowing for excesses. Claims fully absorbed by excess remain in the reported-claim denominator but are excluded from the fitted paid-claim frequency and severity models.

Distinction between simulation and fitted premiums: Module 01 uses a separate internal scale target, `target_mean_technical_premium = 1052`, for its underlying claims calibration. This is not the technical premium used to construct competing quotations or select pricing adjustments. The operative technical premium is $\pi_i = 1.30\hat{\mu}_i$, fitted in Module 02. Its reported mean is \$1,066.83 across the full synthetic portfolio and \$1,070.43 in the 2025 quotation cohort (Section 4.1).

B.2 Baseline Competitor Assumptions

Six stylised competitors are priced relative to the reference insurer's **fitted** technical premium, using $Q_{ik} = \pi_i \exp(\alpha_k + \delta_{ik} + \varepsilon_{ik})$. The log stance α_k is applied before segment adjustments and quote-specific variation. These archetypes are not actual Australian insurers.

Table B.3: Competitor stance and quote availability

ID	Archetype	Log stance α_k	Probability of returning a quote
C01	Value Direct	−0.055	96%
C02	Mainstream Alpha	−0.015	98%
C03	Mainstream Beta	+0.010	96%
C04	Premium Brand	+0.065	90%
C05	Young Driver Specialist	+0.025	92%
C06	Low-Risk Specialist	+0.015	91%

Table B.4: Competitor segment-appetite rules

Competitor	Condition	Log-price adjustment δ_{ik}
Mainstream Beta	Small Passenger segment	−0.020
Mainstream Beta	Sports/Performance segment	+0.020
Premium Brand	Vehicle value $\geq$ \$45,000 and driver age 35–69	−0.020
Premium Brand	Driver age ≤ 24	+0.025
Young Driver Specialist	Driver age ≤ 29	−0.055
Young Driver Specialist	Driver age ≥ 55	+0.018
Low-Risk Specialist	No prior claims and at least three claim-free years	−0.045
Low-Risk Specialist	At least two prior claims	+0.030

These conditions are evaluated in the order specified in Module 03 using `case_when`; **if multiple conditions for the same competitor overlap, the first matching condition applies**, rather than summing the adjustments. The baseline adjustment scale is 1.00. Residual log-price noise has standard deviation **0.045**. Competitor premium multipliers are limited to **0.75**–**1.35** times reference technical premium, and at least **two** competitor offers are retained. The reference insurer's initial commercial multiplier is drawn independently from Uniform $(0.95, 1.05)$.

B.3 Baseline Customer-Demand Assumptions

The two-stage DGP first determines whether a customer purchases insurance through the simulated market and then which insurer is selected, conditional on purchase. The calibration targets are **78%** mean market participation (purchase from any available insurer) and **18%** mean reference insurer conversion. These are assumed simulation targets, not estimated Australian insurance purchasing rates.

Table B.5: Parameters of the baseline demand DGP

Parameter	Baseline value
Mean probability of market purchase	78%
Mean reference conversion probability	18%
Market-participation price sensitivity	0.75
Mean individual conditional-choice price sensitivity	7.00
Conditional-choice price-sensitivity CV	0.35
Rank-salience scale	1.00
Relative-price penalty threshold	1.08
Relative-price threshold penalty strength	0.65
Rank penalty begins beyond	Rank 4
Rank-threshold penalty strength	0.10
Latent taste-shock SD	0.35

Customer-specific price-sensitivity coefficients follow a lognormal distribution. The utility model ranks the premiums of all available insurers, including the reference insurer. Its relative-price ratio uses the cheapest premium across this **complete choice set**. This differs from the reference insurer's market-position variable RP_i, whose denominator is the cheapest **competing** quotation. A calibrated reference insurer utility constant establishes the 18% mean conversion target.

The market-participation linear predictor includes $-0.75\log(\text{customer median quotation/portfolio median quotation})$ and a claims-history adjustment of $0.035\min(\text{prior five-year claim count}, 3)$. Driver-age adjustments are -0.12 for ages up to 24, $+0.02$ for ages 25–34, $+0.10$ for ages 35–69, $+0.02$ for ages 70–79 and -0.06 for ages 80 and above. The intercept is calibrated to the 78% mean market-participation target.

Conditional insurer choice assigns baseline rank-utility bonuses of **0.55**, **0.26** and **0.10** at ranks one, two and three (multiplied by the scenario rank-salience scale). For an alternative with relative price r_{ih} and rank R_{ih}, the nonlinear penalties are

$$Q(r_{ih}) = c_Q\left[\frac{\max\{r_{ih}-r_0,0\}}{0.05}\right]^2, \qquad K(R_{ih}) = c_K[\max\{R_{ih}-R_0,0\}]^2,$$

where r_0, R_0, c_Q and c_K are the thresholds and penalty strengths in Table B.5. The baseline non-price utility is also insurer-specific, as shown below.

Table B.6: Baseline non-price insurer-choice utility adjustments

Insurer archetype	Customer condition	Non-price utility
Reference insurer	Before reference-brand calibration	0.00
Value Direct	All	−0.04
Mainstream Alpha	All	+0.08
Mainstream Beta	All	+0.04
Premium Brand	Vehicle value ≥ \$45,000 and age 35–69	+0.18
Premium Brand	Otherwise	+0.09
Young Driver Specialist	Age ≤ 29	+0.20
Young Driver Specialist	Otherwise	−0.04
Low-Risk Specialist	Zero prior claims and at least three claim-free years	+0.18
Low-Risk Specialist	At least two prior claims	−0.08
Low-Risk Specialist	Otherwise	+0.03

As in the competitor-appetite implementation, the choice utilities are evaluated in order, with the first applicable condition taking precedence. Normal latent preference shocks have baseline standard deviation 0.35. These utility adjustments are stylised behavioural assumptions, not estimated brand preferences from Australian consumers.

B.4 Structural Sensitivity Assumptions

Ten cases were considered in total: the baseline and nine alternatives. Each scenario changes a defined group of assumptions while retaining the remaining baseline structure.

Table B.7: Alternative customer-demand parameters

Parameter	Baseline	Low price sensitivity	Agent-mediated demand	Threshold demand
Participation price sensitivity	0.75	0.50	0.90	0.70
Mean choice price sensitivity	7.0	4.0	10.0	6.0

Parameter	Baseline	Low price sensitivity	Agent-mediated demand	Threshold demand
Choice-sensitivity CV	0.35	0.40	0.25	0.30
Rank-salience scale	1.00	0.60	1.35	0.80
Relative-price threshold	1.08	1.10	1.06	1.05
Price-threshold penalty strength	0.65	0.35	0.95	1.70
Rank threshold	4	5	3	3
Rank-threshold penalty strength	0.10	0.05	0.16	0.34
Latent taste-shock SD	0.35	0.50	0.22	0.30

All demand scenarios retain the calibration targets of 78% market participation and 18% reference insurer conversion. They examine changes in the **shape and strength** of customer price response, rather than changes in the overall starting conversion target. The term *agent-mediated demand* identifies a specified high-price-sensitivity scenario; it does not represent evidence of actual autonomous-agent purchasing behaviour.

Table B.8: Alternative market and pricing assumptions

Scenario	Changes from baseline
High competition	Competitor log-price stances shift **−0.025**; quote-return probabilities increase by **0.04** (subject to probability bounds); log-quote noise SD **0.050**; competitor segment adjustments ×**1.10**.
Low competition	Competitor stances shift **+0.025** and are scaled by **0.90**; quote-return probabilities fall by **0.16**; noise SD **0.035**; segment adjustments ×**0.85**.
Fragmented market	Competitor stance scale **1.55**; availability shift **−0.02**; noise SD **0.080**; segment adjustments ×**1.50**.
Narrow pricing support	Permitted multiplier grid **0.97–1.03** instead of 0.95–1.05, in 0.01 increments.
Strict adequacy	Expected claims-ratio ceiling **78%** rather than 80%.
Wide pricing stress	Permitted multiplier grid **0.90–1.10**, in **0.025** increments; permits extrapolation beyond the reference-pricing range used to generate baseline quote observations.

The baseline targeted optimiser uses premium multipliers from **0.95 to 1.05** in steps of **0.01**, and imposes the following constraints: aggregate quoted rate level from **99.5% to 100.5%**; contribution retention across all customers $\geq 100\%$; contribution retention within each customer group $\geq 90\%$; expected claims ratio $\leq 80\%$; and model-expected conversion retention $\geq 100\%$. At least **50** validation profiles are required per retained segment. Continuous predictors are generally segmented using validation-derived quartiles. If candidate structures differ in validation conversion by no more than **0.0005** in absolute probability (0.05 percentage points), the structure with fewer segments is preferred. Uniform pricing instead maximises model-expected contribution after claims, subject to at least **95%** benchmark conversion and its specified financial constraints; it is not subject to the narrow rate-level restriction.

C Supplementary Results and Model Specifications

This appendix provides the complete comparison of candidate segmentation variables, the selected targeting rule and detailed robustness results that support Section 4. Results labelled **validation** were calculated during model or

strategy selection. Results labelled **test** were calculated after the relevant decisions were fixed. Where specified, test pricing outcomes use the underlying simulated conversion probabilities rather than fitted predictions.

C.1 Complete Comparison of Targeting Variables

Table C.1: Validation comparison of 15 predefined eligible targeting bases

Targeting basis	Segments	Relative conversion uplift	Aggregate quoted rate level	Contribution retention	Minimum group retention
Vehicle-value band	4	19.79%	99.66%	102.28%	90.15%
Vehicle-age band	5	16.92%	99.54%	103.05%	91.33%
Vehicle segment	5	15.48%	99.54%	100.85%	90.08%
Annual-kilometres band	4	11.94%	99.59%	101.93%	90.94%
Unladen-weight band	4	11.66%	99.50%	103.25%	90.14%
Torque band	4	11.62%	99.54%	101.48%	90.11%
Power band	4	11.54%	99.54%	101.59%	90.13%
Claim-free-years band	4	11.09%	99.55%	103.59%	90.17%
Prior-claims band	4	9.98%	99.61%	102.82%	90.19%
Driver-age band	8	9.64%	99.51%	102.87%	90.03%
Licence-years band	4	8.78%	99.63%	101.49%	90.04%
Engine-capacity band	4	8.62%	99.52%	102.82%	90.17%
State	6	8.25%	99.50%	104.14%	90.30%
Voluntary excess	5	7.34%	99.52%	103.36%	90.21%
Transmission	2	7.11%	99.68%	102.41%	100.00%

Each row shows the admissible multipliers giving the highest model-expected conversion for that segmentation variable on the **validation sample**. Conversion uplift and contribution retention are relative to the technical-pricing benchmark and are calculated without consulting test-sample outcomes or underlying simulated conversion probabilities. Segment counts reflect the retained categorical groups or validation-derived numerical bands.

C.2 Selected Vehicle-Value Segments and Multipliers

Table C.2: Baseline validation-selected targeting rule

Vehicle-value group	Validation profiles	Selected multiplier	Model-expected conversion	Segment contribution retention
Q1 (lowest)	501	0.95	31.88%	133.34%
Q2	502	0.99	20.25%	109.40%
Q3	500	1.00	16.48%	100.00%
Q4 (highest)	497	1.01	13.42%	90.15%

The four groups contain all 2,000 validation observations. Vehicle-value cut-offs were determined on validation and remained fixed when assigning test profiles. Segment contribution retention depends jointly on the change in premium and the resulting change in predicted purchases. In particular, the Q1 discount can increase contribution if enough additional business is expected, whereas the Q4 premium increase can reduce it if predicted conversion falls sufficiently. These relationships are consequences of the simulated demand and fitted models, not empirical evidence of causal differences in price sensitivity by vehicle value.

C.3 Structural Sensitivity Results

Table C.3: Complete ten-case targeted-pricing sensitivity results

Scenario	Selected model	Targeting basis	DGP-relative conversion uplift	DGP-evaluated contribution retention	Combined pricing criterion
Baseline	GAM	Vehicle-value band	19.88%	102.85%	Yes
High competition	GAM	Vehicle-value band	19.71%	105.87%	Yes
Low competition	GLM	Vehicle-value band	25.17%	104.45%	Yes
Fragmented market	GAM	Vehicle-value band	29.99%	105.48%	Yes
Low price sensitivity	GLM	Vehicle-age band	1.19%	99.50%	**No**
Agent-mediated demand	GAM	Vehicle-value band	28.90%	110.05%	Yes
Threshold demand	GAM	Vehicle-value band	32.60%	111.43%	Yes
Narrow pricing support	GAM	Vehicle-value band	19.79%	104.23%	Yes
Strict adequacy	GAM	Vehicle-value band	19.88%	102.85%	Yes
Wide pricing stress	GAM	Transmission	9.73%	103.12%	Yes

The combined criterion requires an increase in **underlying expected reference insurer conversion** and contribution retention of at least **100%**, using the underlying simulated conversion probabilities and fitted expected claims costs. The Market-position models improved predictive performance relative to the Own-price GLM in all ten scenarios. Only the low-price-sensitivity scenario failed the combined pricing criterion.

C.4 Monte Carlo Replication Detail

Ten independent runs regenerated Modules 01–06 while retaining Module 00 calibration inputs and baseline structural settings. Model selection and targeted pricing were repeated within each run.

Table C.4: Predictive stability across ten replications

Replication	Selected market-position model	Test log-loss gain over Own-price GLM	DGP Spearman correlation gain
1	GLM	2.59%	0.195
2	GAM	2.46%	0.183
3	GLM	3.43%	0.177
4	GLM	3.04%	0.184

Replication	Selected market-position model	Test log-loss gain over Own-price GLM	DGP Spearman correlation gain
5	GLM	1.71%	0.162
6	GLM	2.63%	0.180
7	GLM	2.27%	0.167
8	GLM	2.47%	0.173
9	GLM	3.07%	0.186
10	GLM	2.47%	0.183

Table C.5: Test pricing stability across ten replications

Replication	DGP-relative conversion uplift	Contribution retention	Expected claims ratio	Aggregate quoted rate level
1	25.40%	103.40%	77.97%	99.51%
2	23.26%	102.44%	77.90%	99.59%
3	24.46%	102.66%	77.92%	99.61%
4	18.54%	102.97%	77.64%	99.60%
5	22.74%	103.29%	78.00%	99.49%
6	25.48%	103.85%	77.98%	99.49%
7	15.57%	104.23%	77.53%	99.52%
8	24.62%	102.95%	77.92%	99.60%
9	23.71%	103.34%	77.97%	99.51%
10	24.49%	103.77%	78.05%	99.38%

Vehicle-value band was selected in all ten replications. Each replication produced positive underlying expected conversion uplift and contribution retention above 100%. The Market-position GAM was selected only once; the Market-position GLM was selected in the other nine runs. The test aggregate quoted rate level fell below 99.5% in replications 5, 6 and 10. The rate-level constraint was imposed during **validation-based selection**, whereas these outcomes were calculated on independently generated test samples. They therefore reflect differences in sample composition and do not demonstrate a violation of the validation optimisation constraint.

C.5 Paired Test Bootstrap

The paired bootstrap draws **5,000** resamples, each containing **2,000 test profiles sampled with replacement**. Each selected customer contributes both their benchmark and targeted-pricing expected outcomes, preserving the within-customer comparison. The selected pricing strategy, fitted model parameters and estimated claims costs remain fixed. The percentile intervals describe variation from the composition of the test sample, conditional on the selected strategy and specified customer-choice DGP; they do not capture uncertainty about the DGP or alternative model selections.

Table C.6: Paired bootstrap summary

Measure	Point estimate	95% percentile interval
Absolute expected conversion uplift	3.63 percentage points	3.34–3.94 pp
Relative expected conversion uplift	19.88%	18.29–21.46%
Contribution retention	102.85%	102.09–103.64%
Targeted expected claims ratio	77.73%	77.65–77.82%
Aggregate quoted rate level	99.60%	99.52–99.68%

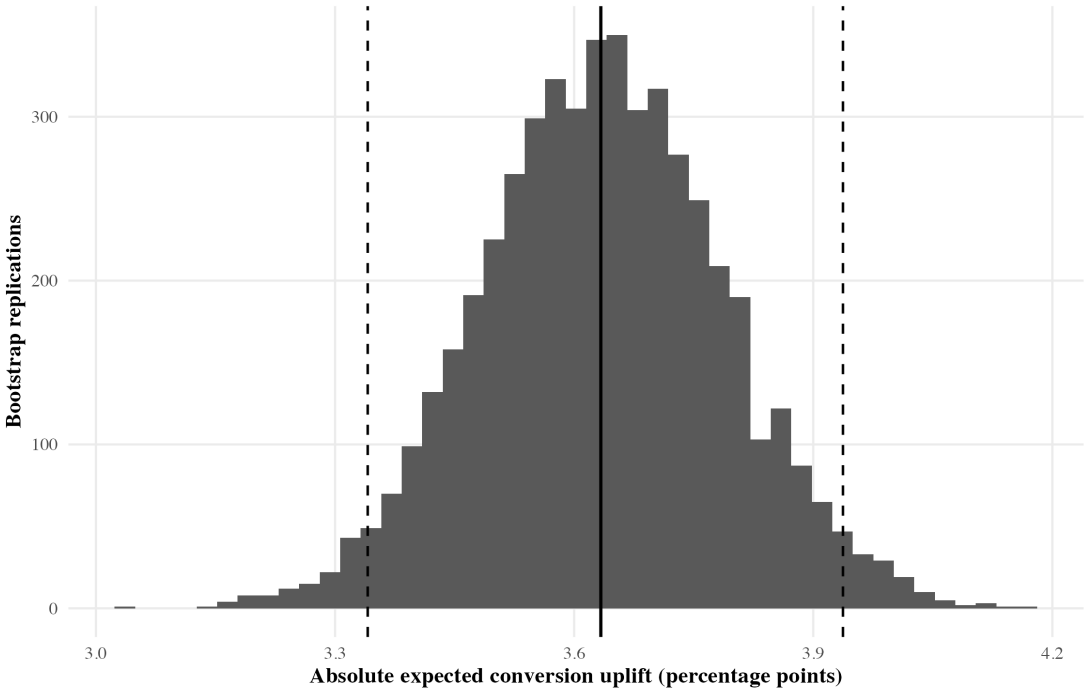


Figure C.1: Bootstrap distribution of the improvement in expected conversion under targeted repricing.

C.6 Model-Specification Register

The following registers summarise the implemented candidate models and selection rules from Modules 02 and 05. They supplement the descriptions in Sections 3.3 and 3.5 and are intended to make the predictor sets and model comparisons explicit.

Table C.7: Technical-pricing candidate specifications and selection

Candidate	Frequency (quasi-Poisson, log link)	Severity (Gamma, log link)
Benchmark	Linear age, licence years, log kilometres; prior claims, State, remoteness	Log vehicle value, excess, vehicle segment
Flexible	Natural splines: age (df 4), licence years (df 3), log kilometres (df 3); prior claims, State, remoteness, segment	Natural splines: log vehicle value (df 4), vehicle age (df 3); excess, vehicle segment
Flexible + motive power	Frequency flexible + motive power	Severity flexible + motive power
Flexible + physical specifications	Frequency flexible + motive power, transmission and physical vehicle specifications	Severity flexible + motive power, transmission and physical vehicle specifications
Selected on 2024	**Flexible + motive power**	**Flexible**

Additional physical-vehicle candidates include engine capacity, power, torque and unladen weight (each with a natural spline of three degrees of freedom), together with cylinder count and transmission. Frequency estimation uses all policies and includes the exposure offset $\log(E_i)$. Severity estimation is restricted to policies with at least one paid claim, weighted by the number of such claims. Model specifications are selected using 2024 validation deviance, refitted on 2021–2024 and adjusted by development-period balancing factors before 2025 evaluation.

Table C.8: Conversion-model information sets and flexibility

Candidate	Distinguishing variables relative to preceding information set
Technical-only GLM	Driver age, licence years, prior-claims band, claim-free years, State, remoteness, vehicle segment, motive power, vehicle age, log vehicle value, log annual kilometres, voluntary excess and log technical premium
Own-price GLM	Technical-only + $\log(P_i/\pi_i)$
Relative-price GLM	Own-price + log reference/cheapest-competitor relative price
Rank GLM	Own-price + reference rank percentile (**rather than also inheriting Relative-price GLM**)
Market-position GLM	Own-price + log relative price to cheapest, rank percentile, excess over market-price threshold, log price relative to market median, log competitor-price spread and centred number of competitor quotes
Market-position GAM	Same broad explanatory set as Market-position GLM, with selected continuous terms represented by cubic regression smooths

The Rank and Relative-price GLMs are **parallel extensions** of the Own-price GLM, rather than stages of one sequentially nested model. In the GLMs, natural cubic splines are used for driver age (4 degrees of freedom), licence experience (3), consecutive claim-free years (3) and vehicle age (3); the remaining transformed continuous terms are entered linearly. The GAM replaces selected linear or spline effects with cubic regression smooths: driver age ($k = 5$), licence experience ($k = 4$), claim-free years ($k = 4$), vehicle age ($k = 4$), log vehicle value ($k = 5$), log annual kilometres ($k = 5$), log technical premium ($k = 5$), log reference-to-technical premium ($k = 5$), log relative price to the cheapest competitor ($k = 6$), rank percentile ($k = 5$), log relative price to market median ($k = 5$) and log competitor-price spread ($k = 4$). Categorical and threshold effects remain parametric. The GAM is selected over the Market-position GLM only when validation log loss improves by at least **0.5%** and the calibration slope is between **0.70 and 1.30**.

D Computational Implementation and Reproducibility

D.1 Module Structure

The experiment is implemented as a sequence of R Markdown modules. Each stage reads the required upstream inputs, applies the specified procedures and writes intermediate datasets, diagnostic summaries and validation checks. This structure separates data generation, model estimation, pricing selection and evaluation.

Table D.1: Computational modules

Module	Main task
00	Prepare and validate Australian empirical calibration inputs
01	Generate the five-cohort synthetic motor portfolio, claim records and withheld latent claims truth
02	Fit/select technical frequency and severity models; estimate 2025 fitted technical premiums
03	Generate reference and competitor quotations, quote availability, competitive-position variables and independent conversion split
04	Generate market participation, insurer choice, observed conversion and withheld DGP demand probabilities

Module	Main task
05	Fit/select logistic GLMs and GAM; evaluate validation and test predictions
06	Select constrained premium adjustments on validation and assess the decision on the test sample against the DGP
06B	Paired bootstrap of the selected test pricing comparison, holding the decision fixed
07	Structural sensitivity analysis, changing one related scenario group at a time
08	Ten full-pipeline Monte Carlo seed replications
09	Dissertation figures and supplementary exhibits

Module 00 prepares the inputs required by Module 01. The baseline experiment then proceeds sequentially through Modules 01–06. Bootstrap, structural-sensitivity and Monte Carlo calculations are conducted after the required baseline outputs are available; Module 09 assembles the dissertation exhibits. Module 06B implements the paired-bootstrap analysis and is provided with the accompanying computational materials.

D.2 Information Controls

The analysis separates the underlying simulation relationships from information used for estimation and selection. The general sequence is: generate claims and demand outcomes, restrict access to the underlying probabilities, fit and validate models, select the pricing decision, hold that decision fixed, and evaluate it using the underlying simulated customer-choice mechanism. The principal information controls are:

1. Technical models are trained on 2021–2023, selected on 2024 and refitted on 2021–2024. The 2025 cohort remains unused during technical model development and selection.
2. A separate 6,000/2,000/2,000 train/validation/test partition is assigned to the 2025 quote profiles before conversion outcomes are generated.
3. Conversion-model selection, probability recalibration and targeted pricing are undertaken without using test conversion outcomes or DGP-expected conversion probabilities for strategy selection.
4. Market-position predictors are reconstructed for each counterfactual reference premium, while competitor quotations remain fixed.
5. The selected targeting variable, segment cut-points and premium multipliers are frozen before independent test evaluation using known DGP conversion probabilities.
6. Validation checks assess required output columns, calibration/consistency conditions, model-prediction reconstruction and pricing constraints.

The technical-pricing cohort split and the conversion-model split serve different purposes. The aggregate quoted rate level may differ between validation and test because the two samples contain different policy characteristics and vehicle-value distributions; a test level outside the validation bounds is not itself evidence that the optimisation constraint failed during selection.

D.3 Code, Inputs and Reproducibility Requirements

Reproducing the analysis requires the final R Markdown modules, their supporting functions in `R/helpers.R`, and the calibration datasets used by Module 00. The empirical dependencies include BITRE registration and licence-holder data, ABS geographic, Census, vehicle-use and price-index information, vehicle-classification inputs, CarExpert vehicle specifications and RACQ vehicle-value references.

The accompanying computational materials contain the final R Markdown modules, required supporting functions and empirical calibration inputs. The saved outputs and fixed random-number seeds document the executed analyses, while the supplied execution order and session information record the software environment needed to reproduce the reported results. The source files should be run in the dependency order summarised in Table D.1